FINANCIAL ENGINEERING PROJECT

Undertow

Making Concentrated Liquidity Pay

Risk-Sensitive Reinforcement Learning on Uniswap v3

Thaker Rahul Dilipkumar

Student ID: A0294733W

Email: e1348871@u.nus.edu

Risk Management Institute

National University of Singapore

Itemized workload	**Hours**
Data collection, ingestion and validation	45
Literature review	35
Programming: pipeline, simulator, exact engine, RL environment, evaluation, tests, report tooling	110
Experiments, analysis, report writing and review	70
Total workload	**260**

Original contribution. This project has the following original contributions:

- An exact, parity-checked testbed for concentrated-liquidity research: an integer fee-growth replay engine, a fast training simulator, and a harness that bounds their fee drift to about 10^{-7} USDC per step on recorded data.
- A joint risk-sensitive reinforcement-learning formulation in which one policy chooses the band width and the hedge ratio, with CVaR used both as a training penalty and as the checkpoint-selection statistic across five seeds.
- The first measurement at the level of a policy, to our knowledge, of learned hedge ratios in concentrated-liquidity provision, replayed against recorded on-chain events with gas, fees and slippage applied to both the liquidity and the hedge leg.
- An empirical active/passive boundary across two fee tiers, with an analytical loss-versus-rebalancing overlay, showing where the fee tier flips the passive sign and why recentring is dominated on the primary pool.
- A diagnostic account of why a standard agent fails on this task, which interface repairs restore learning, and an ablation showing the regime input buys tail protection rather than return.

Undertow: Making Concentrated Liquidity Pay

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Abstract

Providing liquidity on Uniswap v3 looks passive, but it is not. A liquidity provider (LP) chooses a price range, earns fees while the price stays inside it, and loses when informed flow moves the price against the position. On many large pools those losses exceed the fees earned. The LP must decide when to recentre the band, how wide to keep it and how much to hedge. Each choice costs gas, fees or slippage.

We study those decisions as a risk-sensitive reinforcement learning problem on three years of recorded on-chain trades from the 0.05 % (500 pips) ETH/USDC pool, with 2024 held out and frictions charged. The study asks where fees fall short of losses, why a standard learner fails, whether one policy can manage the band and the hedge jointly, and where active management pays.

Fees fail to cover impermanent loss at any width on the primary pool. The 0.30 % tier turns the wide bands net positive, while rebalancing loses on both. A standard learner collapsed into churn; after repairs, five seeds learned to deploy one band and hold. The mean learned policy fails the pre-registered Sharpe test against the frozen best static width (-0.73, CI $[-1.36, -0.04]$) and earns less, but it cuts CVaR relative to the unhedged rules (+23.7 to +34.2). A delta-matched hedge cuts the tail to -3.15 from -59.65 for 2 731 per window. The best seed ties the static width on Sharpe and leads on tail risk. Active management buys insurance in stress and pays for it in calm.

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I Introduction

Automated market makers allow any participant to earn trading fees without quoting prices by hand. Uniswap v3 refined the offer: a liquidity provider (LP) selects the price range in which it is willing to trade, so a deposit of 100 000 USDC earns the fee share of a larger full-range position while the price stays inside the chosen band.[1] Narrowing the band raises the fee earned per dollar of capital, and that prospect drew liquidity into v3.

The promise hides a cost. Liquidity quoted away from the market price is an offer that arbitrageurs accept, and such trades push the pool toward the market price while leaving the LP holding more of the falling asset. Because the loss is a transfer rather than a fluctuation, it does not reverse when the price returns. Milionis et al. named this adverse selection cost loss-versus-rebalancing (LVR) and showed that a full-range position loses value at a rate proportional to the square of volatility.[2] Fritsch and Canidio measured realised arbitrage losses on large pools and found that they exceed the fees those pools earn,[3] and Loesch et al. showed that concentration magnifies the loss once the price leaves the band.[4] Fee income alone is not the LP's return.

An LP who seeks more than a passive range faces three decisions that interact, so they cannot be made independently. When to recentre trades fee income against gas and slippage; how wide to set the next band trades fee density against the time until an edge; how much of the position to hedge trades directional risk against funding and perp fees. These choices arrive as costly impulses under uncertainty about volatility, order flow and regime. The objective is not the highest fee income but the highest return per unit of risk, which places the problem within risk-sensitive impulse control rather

than static portfolio choice.[5]

Recent work addresses parts of the problem. Zhang et al. train a hedged agent for v3 provision, but the hedge executes without cost or funding and the policy sees no regime information.[6] ZeroSwap learns to set a market maker's spread so that quoted prices track an external reference, a different lever from range placement.[7] RAmmStein derives regime-dependent rebalancing thresholds and reports fewer rebalances for the same net return, but its net metric excludes impermanent loss and it works with off-chain data for a single pair.[5] Chionas et al. study risk aversion in concentrated liquidity inside a simulation and leave out on-chain data and the hedge.[8] None of these places timing, width and hedge decisions in a single risk-sensitive policy, charges the frictions on both the liquidity and hedge legs, and tests the result on recorded on-chain trades against hand-written rules.

This report closes that gap. The evidence base is a testbed built from three years of recorded trades in two real ETH/USDC pools. An exact replay backtester computes fees in the protocol's own integer arithmetic, a fast simulator supports training, and a parity harness keeps the two in agreement on identical inputs. On that testbed we ask four questions.

- **Where do fees fail?** We decompose passive returns into fees, impermanent loss, gas and slippage, and identify the conditions under which the deficit persists.
- **Why did the first policies fail?** We document the collapse of a standard policy on this task, trace it to three interface problems, and quantify what the repairs recover.
- **Can one policy do better?** We train a risk-sensitive policy over timing, width and hedging and compare it out of sample with the strongest hand-written rules and a delta-matched hedge.
- **Where does active management pay?** We map the boundary between active and passive provision across volatility, fee tier, gas and capital.

The failures of the first policies are part of the result, not a footnote. In this domain, what the agent is allowed to see and what it is rewarded for matter more than the choice of learning algorithm. A report that showed only the eventual policy would hide the more transferable lesson.

II How Concentrated Liquidity Works

II.A A position as a bet on a range

Uniswap v3 indexes price by integer ticks, and a concentrated position is a bet on the range where the pool price will stay. An LP picks a lower and an upper tick, and the deposit backs trades only while the price remains between them.[1] In protocol units the price is a raw square root, $s = \sqrt{1.0001^{\text{tick}}}$, which increases with the tick; the human price is $q(s) = 10^{d_1-d_0}/s^2$ USDC per WETH, so a lower tick means a higher ETH price. Inside a band $[s_a, s_b]$ the position holds the token amounts

$$x(s) = \frac{L}{10^{d_0}}\left(\frac{1}{s} - \frac{1}{s_b}\right), \tag{1}$$

$$y(s) = \frac{L}{10^{d_1}}\left(s - s_a\right), \tag{2}$$

and its USDC value is therefore

$$V(s) = x(s) + y(s)\,q(s) = \frac{L}{10^{d_0}}\left(\frac{2}{s} - \frac{1}{s_b} - \frac{s_a}{s^2}\right). \tag{3}$$

A fall in the ETH price moves the deposit out of USDC and into WETH; a rise reverses the trade. At an edge of the band the position holds a single asset, and once the price leaves the band it stops trading altogether, so a passive position cannot escape the wrong side of a move.

Fees accrue in proportion to the position's share of active liquidity. Narrowing the band raises that share for a given deposit, which is why concentration is attractive, but it also shortens the band lifetime, the expected time until the price reaches an edge. Fee density and band lifetime therefore pull in opposite directions, and the band width sets the balance between them. That tension shapes the decisions examined throughout this report, and Fig. 1 shows the two effects together.

II.B Two losses, one larger

Liquidity provision is not free, and its cost arrives as two losses. Impermanent loss (IL) is the shortfall of the position against holding the initial deposit. For a full-range position with equally valued legs and price ratio $r = q_t/q_0$,

$$\text{IL}(r) = \frac{2\sqrt{r}}{1+r} - 1, \tag{4}$$

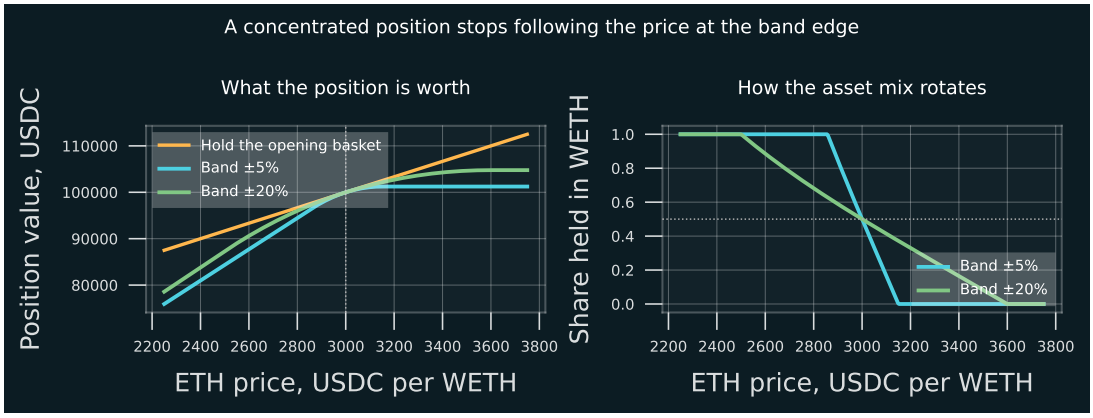


Figure 1: A concentrated position versus a passive holding of the same opening basket. Left: the position tracks the basket while the price sits inside the band and stops following it at the edges, so it gives up the upside of a rally while keeping the downside of a fall. Right: the asset mix rotates from WETH to USDC as the price rises and freezes at the edge. Illustrative parameters: 100 000 USDC entered at 3000 USDC per WETH.

which is second order in the price move: at $r = 1.2$ it is -0.41 %. On v3 the same expression applies only inside the band. Outside it the position is a one-sided holding, so IL depends on where the band was placed.[4]

LVR is the sharper loss. It measures the position against a portfolio that continuously rebalances to hold the same instantaneous exposure, and a second-order expansion of the full-range value shows that the LP falls behind that portfolio at the rate

$$\mathrm{dLVR}_t = \frac{1}{8}\,\sigma^2\,V_t\,\mathrm{d}t, \tag{5}$$

the loss-versus-rebalancing of Milionis et al.[2] Adverse selection therefore grows with the square of volatility and with position size, and fee income must exceed it for passive provision to be viable. Concentration raises fee density but not the fee rate set by the fee tier, so it helps only when the extra fee share outweighs the extra exposure to arbitrage; Section III finds that on many large pools it does not. Figure 2 shows the shape of both losses.

II.C The decisions and the risk

An LP who wants to improve on the one-shot rule, which sets a band once and holds it, faces three decisions, namely when to act, where and how wide to place the next band, and how much market risk to carry. The hedge lever shorts a fraction of the position's ETH delta, and a full short is the delta-matched hedge. The levers interact, so the choice is not separable, and Table 1 states what each lever buys and what it costs. Action is discrete and costly, which places the problem in impulse control,[5]

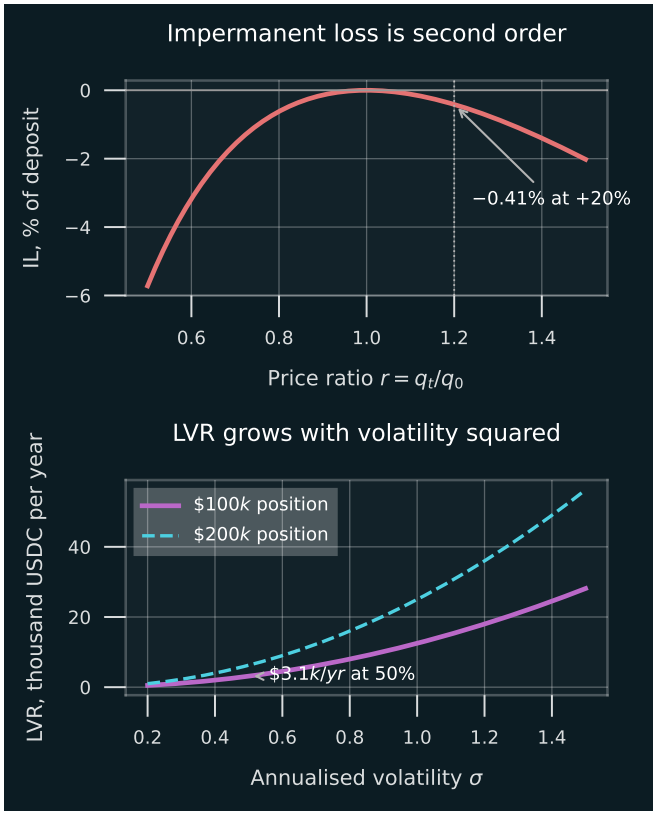


Figure 2: The two losses. Top: impermanent loss against the price move, second order and small for a full-range position. Bottom: the LVR rate against volatility for a 100 000 USDC and a 200 000 USDC position, showing both the quadratic dependence on volatility and the linear dependence on size.

Table 1: The three levers and the trade-off each carries. All returns in the study are measured after the active levers' costs are charged. The liquidity leg pays gas, slippage and impermanent loss, and the hedge leg adds funding and perp fees once the overlay is enabled.

Lever	Choice	Buys	Costs
Timing κ_t	hold or recentre	fee income on a band close to the price	gas and slippage; the fee position restarts
Width w_t	width of the band, in tick spacings	fee density for a given deposit	time until the price reaches an edge
Hedge h_t	short fraction of ETH delta	removal of directional swings	funding and perp trading fees

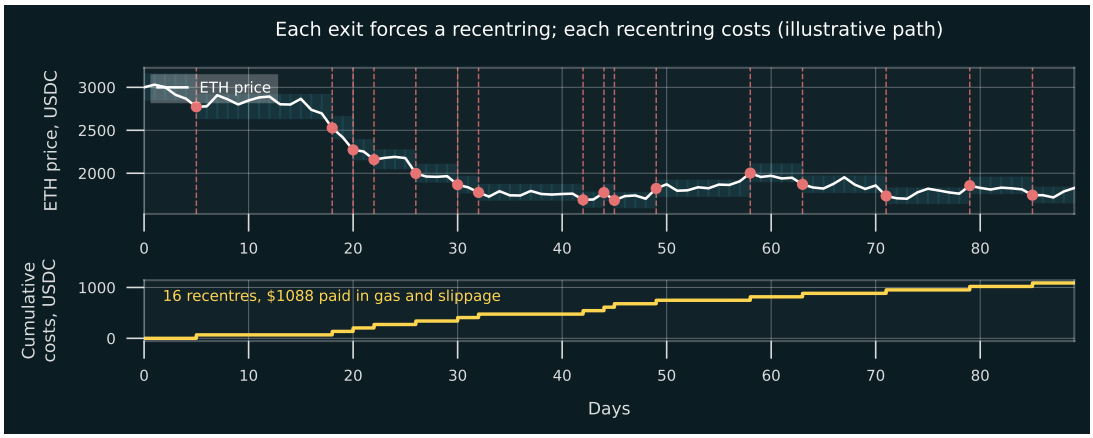


Figure 3: Why acting is a decision rather than a rule. An illustrative price path with a ±5% band: an exit forces a recentring, and each recentring adds gas and slippage to a cost that never falls. The path is illustrative and the costs are simplified; the study charges per-block gas and slippage on the inventory actually swapped.

and the outcome is uncertain, so the objective must be profit per unit of risk rather than profit alone. The primary statistics are the annualised Sharpe ratio and the conditional value-at-risk (CVaR) at the 5 % level; CVaR is coherent and has a convex representation that can be optimised directly.[9] Figure 3 makes the sequence of decisions and costs concrete.

III Related Work

Two literatures bear on the problem. Market making and optimal execution establish how a dealer quotes around a reference price, manages inventory and unwinds a position under uncertainty. The design and economics of automated market makers establish what a liquidity provider earns and loses, and they identify adverse selection as a structural cost rather than an accident of who trades. Reinforcement learning then supplies the decision machinery for a problem of that shape, and its risk-sensitive branch supplies the objective, judging a policy on the tail of its outcomes as well as on their mean. The economic literature prices the problem and the learning literature acts on it; the gap between them is

the provider's decision loop under realistic frictions.

III.A Market making and optimal execution

Classical market making sets quotes around a reference value while managing inventory risk. Avellaneda and Stoikov model a dealer who posts bid and ask quotes against a limit order book, derives a reservation price from inventory and risk aversion, and shows that spreads widen as inventory grows and as the horizon shortens.[10] The same tension drives a much older problem. Almgren and Chriss choose a liquidation schedule that trades price impact against timing risk, and the optimal schedule is the ancestor of the cost-aware rebalancing rule this study uses as a benchmark.[11] Neither model describes an automated market maker. A limit order book dealer chooses quote prices and sizes, while a liquidity provider chooses a price range and a liquidity share; the LP's inventory is a deterministic function of the range, and its loss arises from the concavity of that function rather than from an adverse selection spread.

The reinforcement learning line replaces closed-form quotes with learned policies. Guéant and Manziuk train a policy that quotes across hundreds of corporate bonds at once, sidestepping the curse of dimensionality that makes a per-instrument model impractical.[12] Ganesh et al. build a multi-agent dealer market and show that a learner discovers competitor pricing, shades quotes and manages inventory, with the reward specification deciding much of the resulting behaviour.[13] Zhang et al. study continuous futures with volatility-scaled positions and report gains against time-series momentum after transaction costs.[14] These papers demonstrate that reinforcement learning works in trading settings with realistic frictions, and they supply reward-design lessons that carry into liquidity provision. The state in these studies is an order book and the action is a quote, so range geometry and the losses that make liquidity provision hard stay outside their reach.

III.B Automated market maker design and LP economics

The mechanics of constant-function market makers are well mapped. Angeris et al. prove that a constant-product market tracks the reference price under mild conditions and characterise the deviation when the reference moves away, with agent-based simulations supporting the stability argument.[15] The same authors give conditions under which a constant-function market works as a price oracle,[16] and later work axiomatises the family and shows that a constant-function market admits a unique well-behaved trading function in a general geometric framework.[17,18] Xu et al. systematise the design

space and its trade-offs.[19] This literature explains what a pool does to prices, and it treats the liquidity provider's return as a residual of the mechanism.

That residual is now measured. Capponi and Jia show that automated execution transfers value from liquidity providers to arbitrageurs when exchange rates move, that adoption depends on the pair's usefulness and on the correlation of its legs, and that a more curved pricing curve blunts arbitrage at the cost of trader surplus.[20] Heimbach et al. model the choices a Uniswap v3 provider faces and argue that the design demands far more decisions than earlier automated markets, with fee income and loss exposure moving together.[21] Loesch et al. measure impermanent loss on live v3 positions and show that concentration magnifies it when the price leaves the band.[4] Fritsch and Canidio compare fee income with arbitrage losses on real pools and find that the losses typically exceed the fees earned, with passive full-range positions often outperforming concentrated ones.[3]

The theory of that loss anchors this report. Milionis et al. define loss-versus-rebalancing as the gap between the pool and a continuously rebalanced portfolio, derive the rate $\frac{1}{8}\sigma^2 V$ for a full-range constant-product position, and show that it accrues on any price path, whether or not any trader is informed.[2] Fees change the threshold rather than the mechanism: with a proportional fee, arbitrage only pays once the pool price deviates by more than the fee, and the expected arbitrage profit scales down with the fraction of blocks that present a profitable opportunity.[22] Competition among permissionless providers then pushes liquidity past the social optimum, because a provider ignores the adverse-selection cost its entry imposes on the others.[23] Campbell et al. endogenise the fee itself and show that the profit-maximising fee depends on volatility and volume, so the fee that clears an LP's costs is a function of the market state rather than a protocol constant.[24] The empirical and the theoretical strands agree on the shape of the problem: gross fee yield is not an LP's return, and the gap between them is adverse selection.

What this literature does not supply is a decision rule. Its objects are mechanisms, fee schedules and passive positions, and the provider appears as a price taker of the mechanism rather than an agent that chooses when to act, how wide to set the band, and how much of the residual risk to hedge.

III.C Learning for liquidity provision

A smaller literature learns the provider's decisions. Zhang et al. train a hedged double deep Q-network on a v3 environment and report large gains over baselines, but the hedge is assumed frictionless and the

policy sees no regime features.[6] ZeroSwap learns the spread of a market maker so that quoted prices track an external reference, an orthogonal lever to range placement.[7] Chionas et al. treat concentrated liquidity as a risk-aversion problem in simulation and find that higher volatility and gas widen optimal positions and reduce rebalancing, while a risk penalty trims the adverse tail; their analysis uses no on-chain data and no hedge.[8] Xu and Brini use proximal policy optimisation to train a provider that adjusts positions on rolling windows to balance fees against impermanent loss under regime shifts.[25] RAmmStein poses regime-dependent rebalancing thresholds as a problem of impulse control and reports that lazy rebalancing when the regime warrants it improves net return with far fewer rebalances, though its net metric excludes impermanent loss and the study relies on off-chain data for one pair.[5]

These studies establish that a learned provider can beat a hand-written rule inside their respective settings. None of the studies above charges frictions on the liquidity and hedge legs together, evaluates on validated on-chain replay across several seeds with paired intervals, or places the hedge ratio in the same learned action as the range decision.

III.D Risk-sensitive reinforcement learning

Judging a policy on its tail is a distinct problem from maximising its mean. Mihatsch and Neuneier replace the expectation in the Bellman recursion with an exponential utility and show that a single parameter interpolates between risk-seeking and risk-averse behaviour.[26] Tamar et al. extend policy gradients to coherent risk measures generally, using a convex-programming step over the risk envelope,[27] and Chow and Ghavamzadeh derive policy-gradient and actor-critic algorithms for mean-CVaR Markov decision processes, so that the tail becomes a differentiable objective rather than a post-hoc measurement.[28] Bühler et al. bring the same machinery to hedging under frictions and risk limits, and show that a rich enough set of learned strategies approximates the optimal solution.[29] The measures themselves rest on Artzner et al.'s coherence axioms[30] and on the convex representation of Rockafellar and Uryasev.[9]

This study uses those results in two places. The reward subtracts a multiple of a trailing empirical expected shortfall, so the policy's gradient reflects the tail on which it is evaluated, and the checkpoint selection ranks candidates by a validation Sharpe computed on a held-out slice of the training lane. The second choice responds directly to the failure documented in Section VII, where selection on training return chose an over-hedged policy that lost out of sample.

Table 2: The closest studies and what each leaves open. "Decisions" lists what the policy or model chooses; "Evidence" distinguishes theory, simulation and recorded data.

Study	Decisions	Evidence	What it leaves open
Avellaneda and Stoikov[10]	quote prices and sizes	limit-order-book theory	no range geometry, no pool mechanism
Milionis et al.[2,22]	none (loss accounting)	AMM theory	no provider decision rule
Heimbach et al.[21]	liquidity allocation	model and pool data	static choices, no learned policy
Fritsch and Canidio[3]	none (measurement)	on-chain pools	passive positions only
Zhang et al.[6]	range and rebalancing	simulated v3	frictionless hedge, no regimes
ZeroSwap[7]	quoted spread	simulated market	spread, not range or hedge
Chionas et al.[8]	range and rebalancing	simulated v3	no on-chain data, no hedge
Xu and Brini[25]	active range management	rolling v3 windows	no hedge, no pre-registered risk test
RAmmStein[5]	rebalancing thresholds	off-chain, one pair	net metric excludes impermanent loss
This study	width and hedge ratio	recorded on-chain replay	single pair, one evaluation year

III.E Learning methods and tooling

The optimisation machinery is standard, and deliberately so. Reinforcement learning solves sequential decisions from interaction data without a model;[31] proximal policy optimisation (PPO) supplies a clipped surrogate objective that is stable across updates;[32] generalised advantage estimation controls the variance of its advantage estimates.[33] The study uses the reference implementations of both through Stable-Baselines3 on the Gymnasium interface,[34,35] which keeps the experiments reproducible without making the algorithms part of the contribution. The one interface that is not standard is the action space. The policy emits a factored action whose first component is a range choice and whose second is a hedge ratio, a discrete multi-head space rather than a single categorical output.

III.F The gap

Table 2 places the closest studies beside this one, and two patterns emerge. The economic literature prices the LP's losses but leaves the decisions open. The learning literature makes the decisions but evaluates them in simulation, on one leg, or without the frictions that decide the sign of the result. This study trains a single learned policy over width and hedge on recorded on-chain events, charges gas, slippage, fees and funding on both legs, and reports the pre-registered risk comparison against the one-shot rules and a delta-matched hedge, across five seeds with paired intervals.

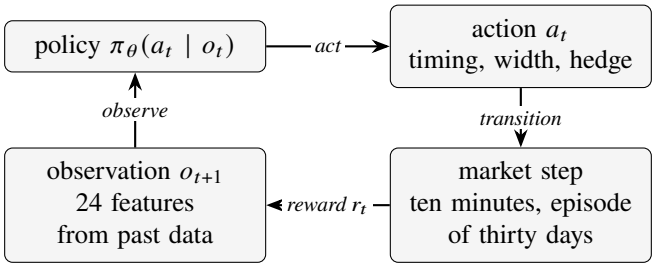


Figure 4: One cycle of the LP decision process. The policy acts every ten minutes; a decision to rebalance moves the band and charges gas and slippage, while a decision to hold leaves the position where it is. Rewards accumulate over an episode of thirty days against the fixed opening basket.

IV The Liquidity Provider's Decision Problem

We model the LP's problem as a risk-sensitive Markov decision process with impulse control. The formalism follows from the economics. A rebalance is a discrete intervention that costs gas and slippage rather than a continuous control, and the volatility regime and order flow that drive the payoff are latent. Decisions are taken every ten minutes over an episode of thirty days, and the agent maps a fixed observation to a choice between holding and acting. The reward measures the strategy's value against the fixed opening basket held passively, net of the costs of the liquidity and hedge legs. Because the regime and the flow are never observed directly, the process is partially observed, and the observation supplies the history on which a decision can condition. Figure 4 shows one decision cycle.

IV.A State

The observation contains 24 features, computed from information available at the decision time. Table 3 groups them by source. Band geometry belongs in the observation because the action space is written in tick spacings relative to the current tick. The distance to an edge sets the time over which the band can earn fees, so a policy that cannot see how close the price sits to an edge cannot time a rebalance. Heavy-tailed features such as price, liquidity and gas are compressed with a signed logarithm, which brings the dimensions onto a common scale for the network.

IV.B Action

At each decision the agent selects a factored action

$$a_t = (\kappa_t,\ c_t,\ w_t,\ h_t), \tag{6}$$

Table 3: The 24 observation features, grouped by source. Every field is computed from information available at the decision time, so an observation at time t cannot contain a bar from $t + 1$.

Group	Features	n
Market	price, sqrt price, tick, 1 h and 24 h log returns, realised 24 h volatility	6
Position	in-range flag, position value, two uncollected fee balances, IL against the fixed basket	5
Pool	active liquidity, fee tier	2
Costs	recent gas price, recent ETH reference price	2
Regime	regime label (bull, bear, sideways, high volatility)	1
Time	step in episode, steps remaining	2
Band	distance to each band edge and band width, in tick spacings	3
Hedge	short ratio, hedge equity, recent funding rate	3
	Total	**24**

Table 4: The four action heads. Widths are given as half the band width, in tick spacings. The general grid holds ten widths; the study's configuration restricts the width head to three arms and the timing head to a single deployment.

Head	Symbol	Options	Meaning
Rebalance	κ_t	hold, act	spend gas and slippage to move the band, or leave it
Centre	c_t	$-4, \ldots, 4$ spacings	shift the centre of the new band around the current tick
Width	w_t	$\{1, 2, 5, 10, 25, 50, 100, 250, 500, 1000\}$ spacings	half of the new band's width
Hedge	h_t	$\{0, 0.25, 0.5, 0.75, 1.0\}$	share of the LP's ETH delta that the short offsets

whose four heads are described in Table 4. Factoring the action keeps the economics of the levers visible in the representation. The study learns the width and hedge heads, while the timing head is exercised once, at deployment, and the centre stays at the current tick. The restriction follows from what the first experiments measured. On this pool the cost of moving a band exceeds the fee that a move can recover, so the timing decision collapses to the choice of whether to deploy at all. Section V documents the diagnostic runs behind the restriction, and Section VII evaluates the learned policy under it.

The width ladder used for the study holds three arms that bracket the economic trade-off. The narrow arm sits near the typical daily move, where fee density is highest and the price is most likely to reach an edge. The middle arm is just wider than the widest hand-written band. The wide arm approaches the full range, where fee density is low but the band survives long stretches of price travel. Together with the hold option these give the liquidity head four choices, and the hedge head multiplies them by five ratios, so a full action is one of 4×5 pairs. The interface also admits the wider timing and centre grids, which remain available for the workstream on active timing.

Table 5: The terms of the reward, in token0 (USDC) units before the $1/W_0$ scaling. Gas and slippage enter as positive magnitudes and carry a minus sign in the reward.

Term	Meaning	Sign
fees_t	fee income earned in the step	+
gas_t	gas for mints, burns, collects and rebalancing swaps	−
slip_t	slippage on the inventory actually swapped	−
$\Delta(V_t - H_t)$	change in position value against the fixed opening basket	±
π_t	mark PnL of the short perpetual	±
ϕ_t	funding paid or received on the short	±
τ_t	perp taker cost of adjusting the short	≤ 0
$\lambda\,\hat{c}_t$	penalty on the trailing expected shortfall of per-step losses, $\lambda = 0.05$ in the study	−

IV.C Reward

A rebalance converts inventory and pays for the privilege, so the reward charges the move and credits the fees the new band earns. Writing V_t for the position value, H_t for the fixed value of the opening deposit held passively, and the hedge cash flows as mark PnL π_t, funding ϕ_t and perp cost $\tau_t \leq 0$, the reward is

$$r_t = \frac{1}{W_0}\Big(\text{fees}_t - \text{gas}_t - \text{slip}_t + \Delta(V_t - H_t) + \pi_t + \phi_t + \tau_t - \lambda\,\hat{c}_t\Big), \tag{7}$$

whose terms are defined in Table 5. Dividing by the initial capital W_0 keeps value targets at a workable scale without hiding any cost. The benchmark is the *fixed* opening basket. Because H_t is never rebased at a rebalance, the episode sum equals the excess over holding that basket, which is the quantity the evaluation reports. Measuring against a rebased entry price would instead credit the agent for value that the reported statistic does not count.

IV.D Risk and benchmarks

The learning objective is the discounted return with a penalty on the recent tail of per-step losses. The risk term in Eq. (7) subtracts λ times the recent 5 % expected shortfall of per-step equity changes, measured over the last trading day, so deeper tail losses cost more reward. The penalty is computed on the joint equity of the liquidity and hedge legs, so the hedge ratio can move the quantity the policy is asked to control. The default configuration disables the term ($\lambda = 0$); the study's configuration sets $\lambda = 0.05$, small next to a full tail loss. The evaluation reports the same quantity the penalty controls.

With losses $L = -G$ defined on the episode return G, the conditional value-at-risk at level α is

$$\mathrm{CVaR}_\alpha(L) = \min_{\nu \in \mathbb{R}} \left\{ \nu + \frac{1}{1-\alpha} \mathbb{E}\left[(L-\nu)^+\right] \right\}, \tag{8}$$

the Rockafellar–Uryasev representation,[9] which we report at $\alpha = 0.95$. Evaluation therefore reports the annualised Sharpe ratio and CVaR at 5 % as primary statistics, with raw PnL and excess over HODL as secondary ones. Comparisons use paired per-window differences and confidence intervals across seeds.

The hedge is a short perpetual position sized against the LP's ETH exposure. Its mark PnL over a step is the negative product of the short notional and the price change, plus funding and the taker cost of adjusting the position. Three reference points make the hedge leg interpretable. The unhedged LP carries the full directional exposure, the policy chooses a hedge ratio step by step, and a hedged HODL portfolio holds a delta-matched short. The third is the benchmark that prevents a claim of success against a weaker opponent. A policy that only removes directional risk must still beat a passive portfolio with the same risk profile. The overlay is enabled in the study's configuration, and the hedge ratio is a learned decision rather than a hand-set constant.

V Building a Trustworthy Testbed

The decision problem framed in Section IV needs a world in which the LP's costs are measured rather than assumed. Such a world must be exact enough that a reported fee or loss is the real one, and fast enough for a learned policy to make millions of decisions. No single engine delivers both, so the testbed runs two and checks one against the other. Constructing that instrument is part of the contribution, because a result is only as trustworthy as the recording and the arithmetic beneath it. Figure 5 shows the flow.

V.A The record

The record begins with the raw logs of two real ETH/USDC pools. The 0.05 % pool is the primary pool, and the 0.30 % pool provides a fee-tier contrast. The pipeline turns those logs into one ordered event tape of swaps, mints, burns and collects, then joins three auxiliary feeds to it, namely one-minute reference prices, per-block gas prices and regime labels computed from thresholds fixed in advance.

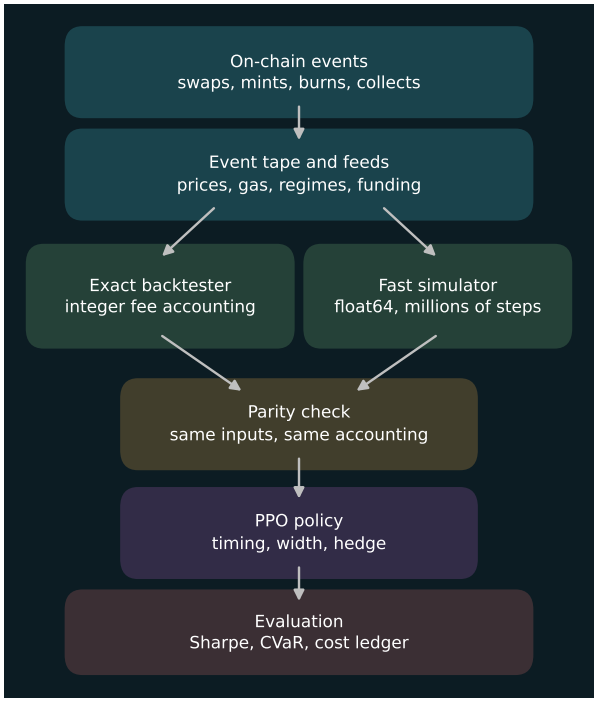


Figure 5: The testbed. Recorded events feed one tape; two engines consume it, an exact backtester for ground truth and a fast simulator for training; the parity check reconciles them before a policy is trusted.

A fourth feed, the funding rate of a perpetual margined in USDT, prices the hedge. The window runs from January 2022 to December 2024, with 2022–2023 for training and 2024 held out. The primary pool's training tape alone holds about 4.86 million events, which is why speed matters later. Reads pass through one interface that enforces block ordering and the train/eval boundary, so an observation at time t cannot accidentally see a bar from $t + 1$.

V.B The exact engine

The exact backtester replays a frozen policy against the tape in the protocol's own integer arithmetic. Fees come from the pool's recorded fee-growth accumulators, exactly as the contract computes them. Liquidity events update an external tick lattice seeded from a boundary snapshot, and swaps are apportioned across the ticks they cross. The ledger therefore satisfies its accounting identity exactly. Fees plus impermanent loss, minus gas and slippage, equals the strategy's excess over holding the opening deposit. Because the interface exposes no way to update the policy, a backtested run is frozen by construction, and the ledger records the configuration and seed behind it.

V.C Two engines, one check

Training on the exact backtester would be far too slow, so the fast simulator mirrors the same economics in floating point. Its agent is marginal. The position never moves the price, and its fee share is the fraction of the active liquidity it supplies. The simulator can run on the recorded events or on a calibrated regime-switching jump process fitted to the training years, and it executes a decision at the price observed before the interval advances, as a live LP would.

Two numeric worlds are only useful if their disagreement is bounded. The parity harness feeds both engines the same prices, positions and gas series, then compares fees, impermanent loss and PnL under a stated tolerance. The simulator's fee engine was made exact, so the remaining drift on real data is tiny. Mean fee differences sit on the order of 10^{-7} USDC per step per seed, and impermanent loss matches exactly. The harness first exercised only the passive policy, which never rebalances. It now also runs a policy that rebalances, which is the setting the fee and loss results depend on. A separate suite of look-ahead probes attacks the composed system at the train/eval seam, where a leak would be easiest to hide.

V.D The failures that shaped the design

The first two training runs produced policies that did nothing useful. The seeds collapsed to one deterministic action, some holding throughout and others repeating a single rebalance, and the measured advantage barely separated the actions. The collapse was not a tuning problem. Table 6 lists the interface defects behind it and the repair for each. Individually the repairs are small. Together they were the difference between a policy that can learn and one that cannot, which is why the failure is treated as a result rather than an implementation note.

VI Experimental Design

The measurement protocol and the decision rule are fixed before the headline comparison, so the evaluation year cannot shape the design judged on it. That ordering matters most for the frozen comparator, since a static width chosen after seeing 2024 would flatter any policy measured against it. The interface corrections that preceded the final runs are reported as corrections rather than absorbed into the design, because they changed what the policy could observe and what it was rewarded for. Table 7 states the

Table 6: The defects that collapsed the first policies, and the repair for each. The first three are interface problems; the last two surfaced while fixing them.

Defect	Consequence	Repair
Observation had no band geometry	the policy could not tell how close the price was to an edge or how far a recentring would move, so it could not time a rebalance	distances to each edge and the band width, in tick spacings
Reward measured loss against a rolling entry basket	after a rebalance the reward no longer matched the reported metric	excess against the fixed opening basket, never rebased
Episodes replayed one fixed window	the agent never saw a different market	training windows resampled from 2022–2023
Slippage charged on the whole position value	costs overstated against a real swap	slippage on the inventory actually swapped
Fee credit taken from global fee growth	a position could be credited more than the fee actually taken when its liquidity dominated a thin segment	the position's exact share of each segment's LP fee

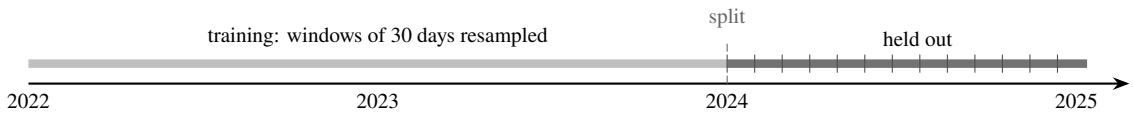


Figure 6: The walk-forward split. Training episodes are windows of thirty days sampled from 2022–2023; evaluation tiles 2024 into windows of thirty days, the last a partial period, and no episode crosses the split.

pools, the split and the training budget, and Fig. 6 shows how the windows sit on the calendar.

VI.A Benchmarks

A learned policy that beats only the weakest rule written by hand has demonstrated nothing, so the ladder in Table 8 includes the opponents that could otherwise account for a positive result. It runs from the one-shot rules through the rebalancing heuristics to the best static width, chosen on the training split and frozen there, and it ends with a delta-matched hedge. That last rung matters because a policy whose advantage is the removal of directional risk would otherwise be measured against opponents that carry that risk. Two reference holdings frame the ladder. A pure buy-and-hold basket holds the opening token mix as tokens and never provides liquidity, which makes it the numeraire for excess over HODL. The hodl rule deploys the opening band before the first decision and never recentres, so it earns fees and pays impermanent loss.

VI.B Metrics

Two primary statistics carry the inference, both computed on the joint equity curve of the liquidity and hedge legs, so a hedge that flattens the tail at a cost in return is scored on the trade rather than on

Table 7: The study protocol and the training configuration. Both pools are USDC/WETH with USDC as token0; prices are quoted in USDC per WETH.

Setting	Value
Primary pool	0.05 % (500 pips), tick spacing 10
Contrast pool	0.30 % (3000 pips), tick spacing 60
Training window	2022-01-01 to 2023-12-31 UTC
Evaluation window	2024-01-01 to 2024-12-31 UTC
Episode	thirty days, one decision every ten minutes
Capital	100 000 USDC per episode
Evaluation tiling	12 windows of 30 days plus a partial final window (13 episodes)
Seeds	5
Training budget	1 000 000 timesteps per seed
Timing	one deployment per episode; the band is held afterwards
Width arms	half-widths of 50, 250 and 1000 tick spacings, plus hold
Hedge head	short ratio in $\{0, 0.25, 0.5, 0.75, 1.0\}$ at every decision
Risk penalty	CVaR at 5 % of the trailing one-day loss, $\lambda = 0.05$
Checkpoint selection	highest validation Sharpe on the last 180 days of training
Algorithm	clipped PPO, $\gamma = 0.9999$, GAE $\lambda = 0.95$, clip 0.15
Network	2 hidden layers of 256 units
Parallel environments	4

Table 8: Benchmark policies. The learned policy is compared with all of them; a positive claim requires beating the best static width and the delta-matched hedged benchmark, not just hodl.

Benchmark	Rule
hodl	Hold the opening band of half-width 120 ticks; never recentre.
passive_narrow	Deploy a ±5 % band once; never recentre.
passive_wide	Deploy a ±20 % band once; never recentre.
full_range_v2	The Uniswap-v2-equivalent full range.
tau_reset	Recentre whenever the price exits the live band.
cost_aware	Recentre at most daily, and only when the estimated fee gain exceeds gas.
Best static width	The ladder width with the best Sharpe on the training split, frozen for test.
Hedged HODL	The opening band held, plus a short perpetual matched to its delta, with funding and perp costs.
Learned policy	PPO over the joint (timing, width, hedge ratio) action space.

one side of it. The annualised Sharpe ratio is the mean of the per-step equity returns divided by their standard deviation, scaled to a year at the ten-minute cadence, and it measures the return per unit of risk. CVaR at the 5 % level is the average of the worst 5 % of per-step equity changes in USDC, the same quantity the reward penalty controls, and it measures the depth of the loss when the market turns. The learned policy is compared with the benchmark ladder through paired per-window differences in both statistics, with bootstrap confidence intervals over the thirteen windows. Because the tail is thin at that sample size, the worst window is reported alongside the intervals. Secondary statistics are raw PnL, excess over HODL, annualised return, Sortino ratio, maximum drawdown, rebalance count, and the cost ledger of fee income, gas, slippage, realised impermanent loss and the hedge flows. Per-regime tables stratify the windows by their dominant label, namely bull, bear, sideways or high volatility.

VI.C Ablations and the boundary

The failure modes were repaired in sequence rather than varied in a factorial design, and the sequence is reported as such. Table 9 lists the design choices the repairs touched, namely band geometry, the reward basket and window sampling, the rebalance interval, the hedge head and the risk penalty. A sequence cannot isolate the marginal effect of a single change, because a repair landed on an interface the next repair could also move, so Section VII reports the progression rather than an attribution. Three targeted probes complete the table and recover part of what the sequence cannot separate. The friction terms are ablated on a one-shot band and on a recentring rule, the rule ladder is repeated on the 0.30 % tier, and an arm with the regime feature frozen is trained across the five seeds.

The boundary question calls for a sweep of realised volatility, volume relative to liquidity, fee tier, gas price and capital, evaluated on replayed windows rather than a fitted process. The choice keeps a boundary point tied to market states that occurred, so the sweep does not add a generator's assumptions to those already carried by the testbed. The completed sweep moves the gas and capital levels and the fee tier, while the realised windows supply the volatility and volume variation the LVR overlay uses. A controlled volatility and volume sweep remains open.

The pre-registered rule awards a positive verdict only if the learned policy's mean Sharpe beats the best static width and the delta-matched hedge with a paired 95 % confidence interval that excludes zero, and only if the policy's CVaR is no worse than the best heuristic's. Both clauses must hold, so tail protection without a return advantage does not pass. Section VII reports where the rule is met and where it is not. Where the evidence is negative, the negative result is the finding.

Table 9: The design choices the repair sequence touched, plus the targeted probes that complete the table. The repair changes were made in sequence rather than varied one at a time, so Section VII measures the sequence of repairs, not the marginal effect of each change; the friction, fee-tier and regime-feature rows are the probes.

Design choice	Ablation	Question it answers
Band geometry	removed from the observation	does the agent need to see the live band to time rebalances?
Window sampling	windows held fixed instead of resampled	how much does resampling the training windows buy?
Hedge	overlay disabled	what does the hedge contribute net of funding?
Risk penalty	toggled	does the risk penalty change the tail without costing Sharpe?
Regime features	frozen to a constant across the five seeds	do the regime labels inform the policy?
Frictions	full model, no gas, no slippage, flat gas, no impermanent loss	which cost drives the result?
Fee tier	repeated on the 0.30 % pool	does a higher fee tier buy a longer viable range?

VII Results

This section addresses the four questions in the order in which they were posed. The results pertain to the 0.05 % ETH/USDC pool and cover thirteen 2024 windows that no policy observed during training. Where a learned policy is involved, results are averaged across five seeds. Statistics are computed on the exact backtester, and the testbed applies its frictions to whichever leg is active.

VII.A Where the fees fail

A passive position collects fees and bears the cost of price movements it cannot follow. Table 12 separates these two components. Fee income declines as the band widens, while impermanent loss declines more rapidly. The liquidity leg is negative across the passive ladder. The deficit narrows as the band widens, but its sign does not change. Even the strongest passive rule finishes below the buy-and-hold basket.

Narrowing the band increases fee income and also increases the magnitude of impermanent loss. The two quantities grow in roughly equal proportion, so their ratio remains approximately constant across the widths reported in Table 12. Concentration therefore leaves the shortfall intact. It changes the variance of the outcome rather than its sign.

The window-level results are consistent with the aggregate pattern. The final column of Table 12 counts negative liquidity windows by rule. This count declines as the band widens but never reaches

Table 10: Friction ablation on the 0.05 % pool over the thirteen evaluation windows. Values are mean PnL in USDC with Sharpe in parentheses. The one-shot rule is passive_wide; the recentring rule is tau_reset. no IL removes the impermanent-loss term. no gas and no slippage remove their cost term. flat gas re-prices each rebalance at a flat 30 gwei base fee plus a 3 % tip at 3000 USDC per ETH.

Variant	one-shot wide	recentring
full	+1 623 (1.53)	−17 680 (−5.81)
no IL	+5 172 (1.65)	+31 224 (12.84)
no gas	+1 659 (1.53)	−8 908 (−3.02)
no slippage	+1 624 (1.53)	−12 359 (−3.91)
flat gas	+1 583 (1.53)	−18 303 (−6.32)

zero. Losses concentrate in windows where the price trends or jumps. In range-bound windows the narrower bands change sign and fee income covers the shortfall. Table 17 reports the same pattern by regime.

Active rebalancing does not remedy the arithmetic and in fact worsens it. The two rebalancing rules in Table 12 collect more fee income than any one-shot rule because they keep the band close to the price. They also incur larger realised losses and execution costs. Both finish further behind the basket than any passive rule. tau_reset loses in all four regimes. cost_aware earns only in bull windows. For this pool and year, fee income does not cover the price risk a position carries. Frequent action adds cost without changing the sign. The next subsection examines whether a learned policy can exploit this structure.

The decisive cost is the one the position cannot avoid. Table 10 removes one cost term at a time from the same ledger. When impermanent loss is removed, the recentring rule reverses from a large loss to a large gain. The one-shot band also gains materially. When gas and slippage are removed, the one-shot band barely changes. The frequently rebalancing rule recovers its execution costs but remains far behind. The one-shot rule loses to the basket because of price travel rather than execution. The recentring rule adds execution costs to a loss it has already increased.

A higher fee tier reverses the sign of the passive arithmetic. Table 11 and Fig. 7 rerun the ladder on the 0.30 % pool. The wide and full-range arms, together with both static widths, finish ahead of the opening basket. A larger fee per unit of volume covers the loss to informed flow. The two rebalancing rules still lose. The contrast indicates what the primary pool would require. The same widths cross from deficit to surplus on a tier with six times the fee per swap.

Table 11: The rule ladder on the 0.30 % (3000 pips) tier, mean over the thirteen evaluation windows. Net is the liquidity leg's excess over the opening basket. The two static rows are the price-equivalent of the primary pool's 250- and 1000-spacing arms at this pool's spacing of 60.

Rule	PnL	Sharpe	Net
hodl opening band	+1 361	2.06	−109
passive_narrow (±5 %)	+1 432	2.89	−38
passive_wide (±20 %)	+1 808	3.33	+338
full_range_v2	+2 188	1.30	+719
static width 42 spacings	+1 893	3.45	+423
static width 167 spacings	+2 204	1.97	+734
tau_reset	−20 702	−15.89	−22 172
cost_aware	−2 101	−0.27	−3 571

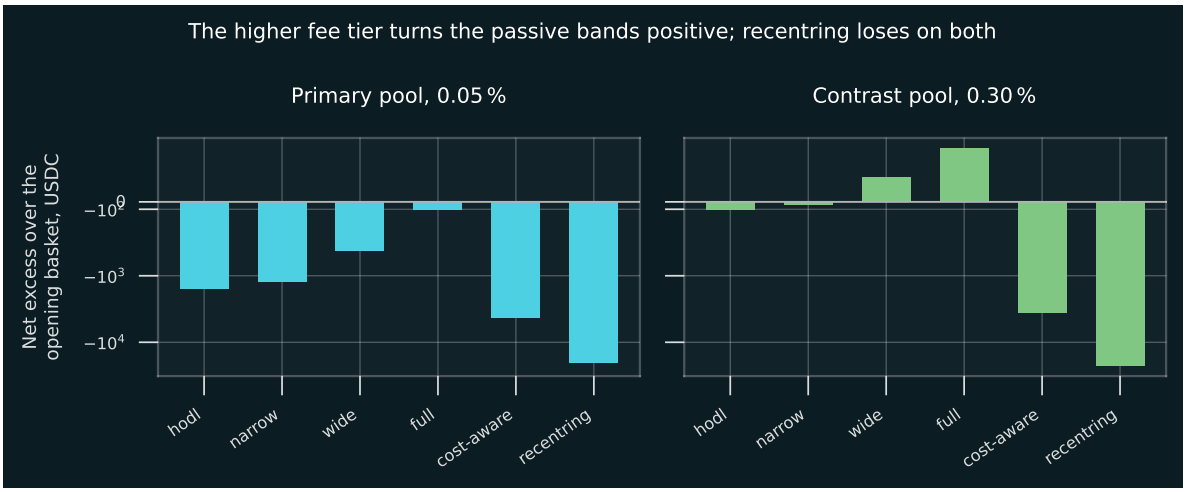


Figure 7: Net excess over the opening basket on the two fee tiers, mean over the thirteen 2024 evaluation windows. The primary pool is the 0.05 % tier. The contrast pool is the 0.30 % tier. Bars above zero beat the passive holding. The vertical axis is symmetric-logarithmic so the recentring rules remain visible.

Table 12: Fee income against impermanent loss on the thirteen evaluation windows. Values are mean per window in USDC. Fees, impermanent loss, gas and slippage are measured against the pure buy-and-hold basket. The liquidity net therefore excludes the basket's own 2288 USDC, and the hedge column is zero for the unhedged rules. The last column counts windows in which the liquidity net is negative.

Policy	Fees	IL	Costs	LP net	Hedge	Net<0
hodl	+4 734	−6 287	0	−1 553	0	6/13
passive_narrow	+4 266	−5 435	−37	−1 206	0	5/13
passive_wide	+2 921	−3 549	−37	−665	0	4/13
full_range_v2	+362	−430	−37	−106	0	3/13
tau_reset	+43 029	−48 904	−14 093	−19 968	0	13/13
cost_aware	+19 833	−22 038	−2 075	−4 280	0	10/13

VII.B Why the first policies failed

The first question was whether the learner could act at all. On the original interface it could not. The timing, centre and width decisions were encoded as 91 actions, and the reward was measured against a basket that moved whenever the band was rebalanced. The trained policies churned. They shifted the band at frequent intervals and paid gas and slippage whenever they moved. Reducing the grid to 28 actions changed the magnitude of the churn but not the behaviour. The collapse was not an optimisation accident. The reward gradient was dominated by the decision to act or hold, and the observation provided no signal about the position of the price within the band.

The repairs were applied in sequence rather than as a factorial ablation. The evidence therefore measures the cumulative effect of the sequence and not the marginal effect of any single change. Three interface defects were corrected first. The observation gained the distance to both band edges and the band width. The reward was measured against the fixed opening basket rather than a basket that moved on rebalance. Training windows were resampled from 2022–2023 rather than replayed from one fixed month. Two further defects emerged during these repairs. Slippage was charged on the whole position value rather than the inventory actually swapped, and fee credit was taken from global fee growth rather than the position's exact share of the occupied liquidity segments. Both were corrected before any result reported here.

Three changes to the learning setup followed. Stabilising the update targets made the five seeds hold for the full episode with zero rebalances and turned a strongly negative return into a mildly negative one. The update remained inside its trust region. The policy had learned to do nothing, which tied hodl. Reducing the action space to a one-shot band choice and lengthening the effective horizon together gave the width decision a usable signal. The first pilot that genuinely selected a band matched the full-range rule's PnL with a higher Sharpe ratio and beat hodl on both PnL and Sharpe.

The hedge head failed in its first joint run. The seeds held the liquidity position for the full episode and collapsed the hedge ratio to a seed-specific constant. The seed selected on the training curve lost out of sample. The objective had never rewarded tail protection, and selection used the training return rather than a validation statistic. The final training run added a penalty on the conditional tail of joint equity and selected checkpoints by validation Sharpe. That run produced the final policy evaluated below.

The regime label is an input rather than a target, but it still shapes what the policy learns. A second arm

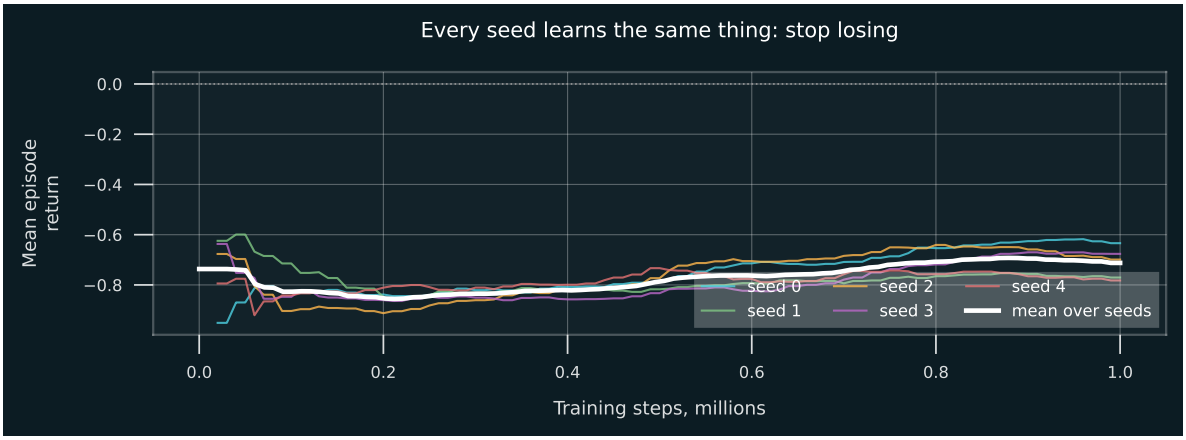


Figure 8: Training reward of the five seeds of the main joint run. The vertical axis is mean episode return against timesteps. The return is measured against the opening basket and normalised by capital. A less negative curve is a policy that has learned to lose less.

trained five seeds with the regime feature frozen to a constant. Its mean Sharpe is numerically higher, but the paired difference is indistinguishable from zero at this sample size. Its mean CVaR is worse, with an interval that excludes zero. The live regime feature therefore buys tail protection rather than return. The ablated seeds diverge more widely. One never deploys, and two widen to the 1000-spacing band. The arm's best validation seed finishes worst out of sample. The selection signal remains weak even for a policy that learns. Appendix F reports the arm in full.

VII.C Whether one policy does better

The pre-registered comparison uses the strongest hand-written rules, the best static width selected on the training split, and a hedged benchmark with a matched delta. Table 13 reports the levels, and Table 14 reports the paired per-window differences on which the decision rule depends.

The frozen comparator provides the first test. On the training split the narrowest of the three width arms had the best Sharpe, so the protocol froze that width. Appendix F lists the training and evaluation scores. The learned mean loses to the frozen width on the evaluation windows. The paired Sharpe difference is -0.733 with an interval of $[-1.355, -0.041]$ that excludes zero. The learned mean also loses to the wide band, with an interval that excludes zero. The Sharpe clause of the pre-registered rule fails.

The tail clause holds against the unhedged benchmarks. Table 13 reports the levels, and Table 14 reports the paired differences. The CVaR differences against unhedged benchmarks are positive, with intervals that exclude zero. The clause fails against the delta-matched hedge, whose paired interval excludes zero

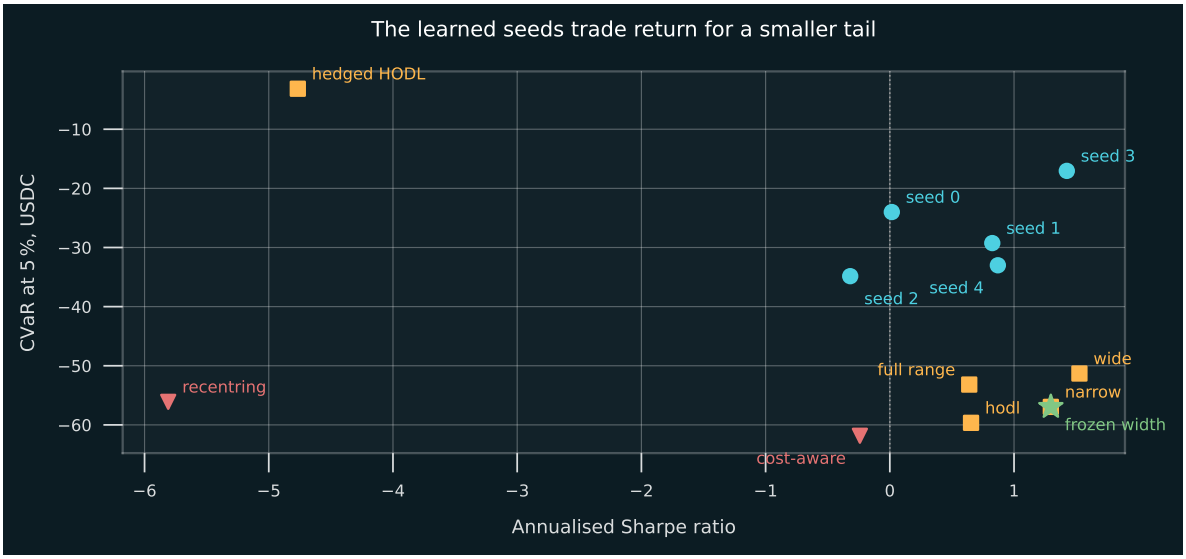


Figure 9: Annualised Sharpe against CVaR at the 5 % level for the rules in the evaluation ladder, mean over the thirteen 2024 windows. Up and to the right is better. The star marks the static width frozen on the training split. The learned seeds sit above the unhedged rules on the tail axis but left of the best static width on the return axis.

in the opposite direction. Both clauses must hold for a positive verdict, so the pre-registered rule yields none. The evidence supports a reduction in tail risk but not a superior return.

Aggregation across seeds conceals the strongest individual policy. The seed with the highest validation Sharpe during training deployed the 250-spacing band throughout the thirteen windows and held it. Its row in Table 13 is the best seed. Its Sharpe is statistically indistinguishable from that of the frozen width, while its CVaR is better with an interval that excludes zero. It beats the delta-matched hedge on Sharpe and loses to it on CVaR. A single seed is not the pre-registered unit, and a seed chosen after observing the evaluation windows cannot support the verdict. This seed did have the best validation Sharpe of the five on the training split, so its selection did not use the evaluation year.

The seeds agree on timing but disagree on width and hedge. A seed deploys once at the start of the episode and holds. Table 15 records one rebalance per episode for the five seeds. Seed zero uses the narrow band. Three seeds use the 250-spacing band throughout the evaluation year. Seed one uses the 250-spacing band in twelve of the thirteen windows. Hedge ratios are partial and differ by seed. The learned policy sets a band and a hedge ratio rather than a timing rule. Its preferred width is the static band with the best Sharpe on the evaluation windows, although that band was not the frozen comparator. Figure 10 shows deployment by seed.

The worst window for the one-shot rules and the learned mean is the single high-volatility window. Table 17 reports its PnL and tail loss by regime. Raw PnL and tail risk rank the policies consistently

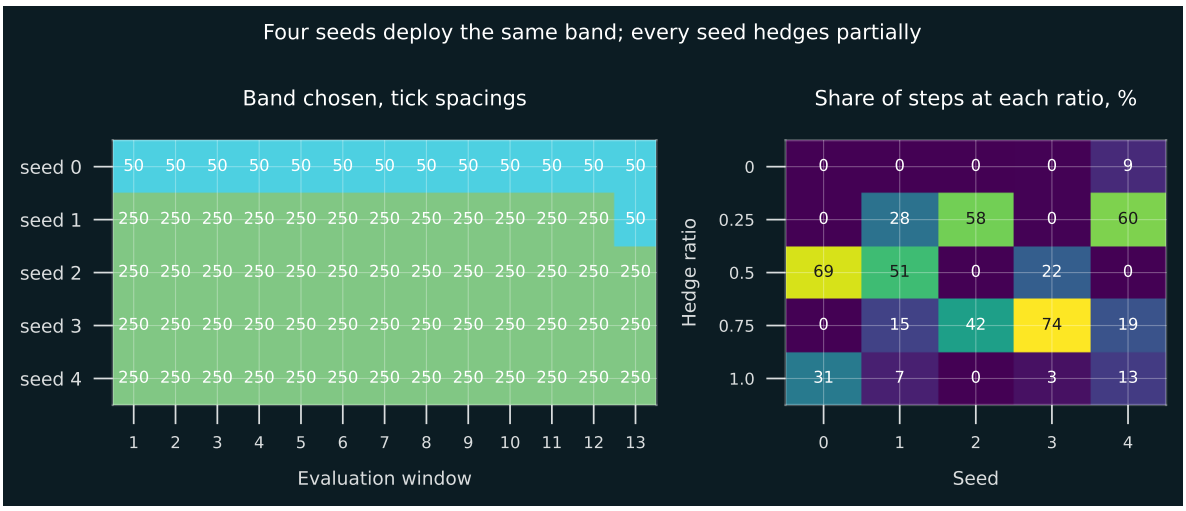


Figure 10: What the learned seeds chose on the thirteen evaluation windows. The left panel shows the band half-width in tick spacings chosen at deployment, with hold where the seed never deployed. The right panel shows the share of decision steps spent at the five short ratios.

Table 13: Performance in the 2024 evaluation on the 0.05 % (500 pips) ETH/USDC pool, mean over the thirteen windows of the 2024 tiling. PnL and excess are USDC per episode. Excess is measured against the pure buy-and-hold basket, the study's HODL numeraire. The row for the learned mean averages the five seeds. The row for the best seed reports the seed with the highest validation Sharpe on the training split. Rebalances counts executed moves per episode. The frozen comparator selected on the training split is the half-width of 50 tick spacings, the passive_narrow row.

Policy	PnL	Excess	Sharpe	CVaR	maxDD	Reb.
Basket	+2 288	0	0.441	−53.81	−9.6 %	0
hodl	+735	−1 553	0.654	−59.65	−12.2 %	0
passive_narrow	+1 082	−1 206	1.296	−56.95	−11.3 %	1
passive_wide	+1 623	−665	1.526	−51.31	−10.1 %	1
full_range_v2	+2 183	−106	0.639	−53.18	−9.7 %	1
tau_reset	−17 680	−19 968	−5.810	−56.10	−21.9 %	122.8
cost_aware	−1 992	−4 280	−0.241	−61.84	−14.0 %	22.0
Hedged hodl	−1 996	−4 284	−4.766	−3.15	−2.7 %	0
Learned mean	−310	−2 599	0.563	−27.63	−6.0 %	1
Best seed	+175	−2 113	1.425	−17.04	−3.2 %	1

across the year. No paired PnL difference against the ladder excludes zero except the comparison with tau_reset, so the return evidence remains inconclusive at this sample size.

VII.D Where active management pays

The regime split provides the final answer. The hedge offers insurance rather than outperformance. Table 17 splits the evaluation windows by dominant label. The counts are uneven by construction. There are five bear windows, five bull windows, two sideways windows and one high-volatility window. The high-volatility row is therefore a case study rather than a distribution.

Table 14: Paired per-window differences, learned mean minus benchmark, with bootstrap 95 % intervals over the thirteen windows. A star marks an interval that excludes zero. A positive Sharpe difference favours the learned policy. A positive CVaR difference means a smaller tail loss than the benchmark. The CVaR column is in USDC.

Benchmark	Sharpe	CVaR
hodl	−0.091 [−0.808, +0.750]	+32.019 [+21.319, +42.230]*
passive_narrow	−0.733 [−1.355, −0.041]*	+29.320 [+19.365, +38.765]*
passive_wide	−0.963 [−1.472, −0.509]*	+23.677 [+18.298, +29.475]*
full_range_v2	−0.076 [−0.930, +0.693]	+25.549 [+17.841, +33.977]*
tau_reset	+6.373 [+4.702, +8.278]*	+28.471 [+21.456, +36.160]*
cost_aware	+0.804 [−0.222, +1.656]	+34.207 [+26.501, +42.092]*
Hedged hodl	+5.329 [+2.550, +7.437]*	−24.481 [−30.423, −18.945]*

Table 15: Deployment by seed on the thirteen evaluation windows. Bands are half-widths in tick spacings, with the number of windows in which the seed chose that band. Hedge ratios are shares of decision steps at each short fraction. All seeds rebalanced exactly once per episode.

Seed	Band (windows)	Hedge ratio (share of steps)
0	50 (13)	0.50: 69.5 %, 1.00: 30.5 %
1	250 (12), 50 (1)	0.50: 51.0 %, 0.25: 27.9 %, 0.75: 14.6 %, 1.00: 6.5 %
2	250 (13)	0.25: 58.2 %, 0.75: 41.8 %
3	250 (13)	0.75: 73.8 %, 0.50: 22.1 %, 1.00: 3.4 %, others: 0.7 %
4	250 (13)	0.25: 59.5 %, 0.75: 18.8 %, 1.00: 12.9 %, 0.00: 8.7 %

In the quiet windows, passive holding performs best. The sideways row of Table 17 shows the one-shot bands collecting fee income while the delta-matched hedge reduces the result. The learned seed retains only part of the available return. Fee income is most reliable when the price does not travel, and in those windows a unit of hedge costs more than it protects.

Trending windows behave differently by direction. In the bull row the full range earns the largest PnL, while the learned seed gives up part of the rally through its short. In the bear row the ordering changes. The wide band gains by holding USDC through the fall. The learned seed has the best risk-adjusted profile on the panel and a bear CVaR far smaller than hodl's. The learned policy's bear-market advantage appears in the tail panel rather than the PnL panel.

The single high-volatility window is where active management earns its premium. The unhedged rules lose heavily. The learned seed loses about a quarter of the hodl loss and carries a much smaller tail. No policy has a positive Sharpe in that window. The delta-matched hedge loses the least. It produces the smallest loss in the stressed window but the largest hedge cost over the year. Widening the band or hedging the delta reduces the loss in stress, although neither choice pays for itself in the quiet windows.

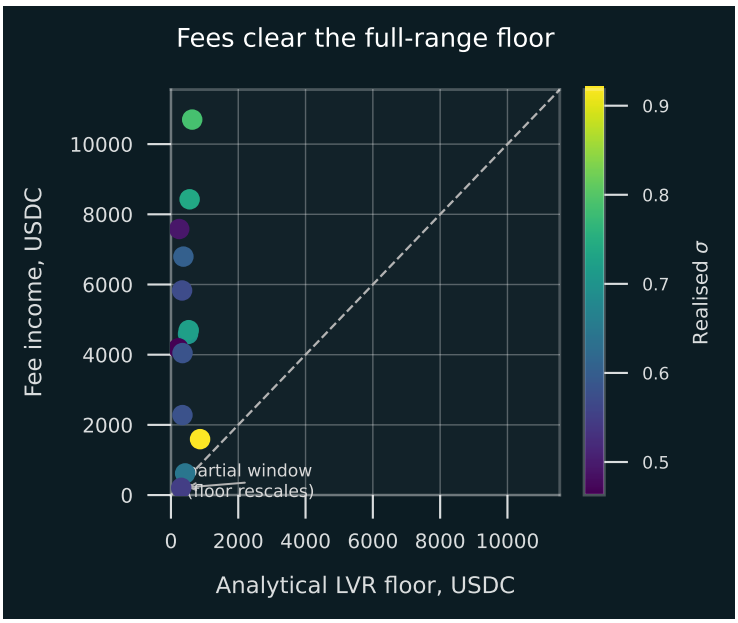


Figure 11: Fee income of the opening band against the analytical full-range LVR floor $0.125\,\sigma^2 Vt$, one point per evaluation window, coloured by the window's realised volatility. The dashed line is where fees exactly cover the floor. The annotated point is the partial final window, whose floor rescales to its length.

The controlled sweep moves along both sides of the boundary. Table 16 reruns the ladder at a quarter and four times the recorded gas price, and at a tenth and ten times the capital. No level turns the recentring rules positive in PnL. `tau_reset` worsens as gas rises. `cost_aware`, whose gate suppresses moves as gas rises, remains small but negative across the gas levels. Cheap execution does not revive timing, and a larger book does not revive it either, because fees and impermanent loss scale with the book while the fixed gas advantage remains small. The learned seed deploys once at all six levels and holds, with a stable Sharpe ratio.

The fee tier is the dimension that reverses the passive sign, and the analytical LVR rate explains why. A liquidity position loses $0.125\,\sigma^2 Vt$ to arbitrage as a floor.[2] The opening band's fee income covers that floor in the full windows, and in the partial final window once the floor is scaled to its shorter length. The thinnest margins occur in the two windows with the least fee income. Fig. 11 plots the comparison. On this fee tier the same widths still lose to the opening basket, and they cross into surplus on the 0.30 % tier. The evaluation year contains one high-volatility window, so the stress row remains a case study. The boundary drawn here holds for this pair, this year and the tested ranges.

Table 16: The boundary sweep on the 0.05 % pool. Values are mean PnL in USDC over the thirteen evaluation windows, with Sharpe in parentheses. The gas levels scale the per-action gas cost. The capital levels scale the book. The last column is the best validation seed of the main run.

Level	hodl	wide	tau_reset	cost_aware	best seed
Gas ×0.25	+735 (0.65)	+1 651 (1.53)	−11 613 (−3.72)	−2 953 (−0.54)	+203 (1.42)
Gas ×1	+735 (0.65)	+1 623 (1.53)	−17 680 (−5.81)	−1 992 (−0.24)	+175 (1.42)
Gas ×4	+735 (0.65)	+1 509 (1.53)	−40 032 (−12.85)	−1 419 (0.01)	+65 (1.42)
Capital 10k	+81 (0.68)	+128 (1.53)	−6 764 (−15.37)	−195 (−0.10)	−13 (1.46)
Capital 100k	+735 (0.65)	+1 623 (1.53)	−17 680 (−5.81)	−1 992 (−0.24)	+175 (1.42)
Capital 1M	+1 384 (0.46)	+16 213 (1.51)	−162 687 (−5.47)	−54 962 (−1.35)	+2 003 (1.40)

Table 17: Mean PnL (top panel) and mean CVaR at 5 % (bottom panel) in USDC by dominant regime label over the 2024 evaluation windows. There are five bear windows, five bull windows, two sideways windows and one high-volatility window. The Width 50 and Width 202 columns are the narrow (±5 %) and wide (±20 %) one-shot bands, whose mean half-widths are 50 and 202 tick spacings. The column for the best seed reports the seed with the highest validation Sharpe on the training split. The single high-volatility window is a case study, not a distribution.

	Mean PnL (USDC)					
Regime (windows)	hodl	Width 50	Width 202	Full range	Delta hedge	Best seed
Bear (5)	+317	+1 772	+2 589	+1 044	−1 707	+527
Bull (5)	+4 335	+4 127	+4 779	+7 379	−1 958	+1 093
Sideways (2)	+4 648	+3 558	+1 693	−550	−3 145	+74
High volatility (1)	−23 000	−22 542	−19 123	−12 642	−1 330	−5 969
	Mean CVaR at 5 % (USDC)					
Bear (5)	−78.3	−74.3	−57.3	−49.2	−2.0	−19.6
Bull (5)	−31.9	−31.2	−37.0	−58.3	−4.1	−12.8
Sideways (2)	−65.0	−59.3	−50.8	−49.6	−4.4	−15.8
High volatility (1)	−94.5	−94.4	−93.9	−54.5	−1.4	−27.8

VIII Discussion

VIII.A The boundary the windows draw

The four answers point to a single boundary that stress conditions define. Passive provision paid in the quiet windows, where fee income is reliable and the unhedged bands keep it. Breadth mattered more than action in the trending windows, because a wide band carries the market exposure that a narrow band sells away. The one high-volatility window reversed the ordering. The unhedged rules lost heavily, while the hedged and wider positions lost less. The learned policy sits on the defensive side of that boundary, and the premium it pays is visible in the calm windows. On tail risk the boundary moves in its favour, while on PnL it stays with the passive bands. Table 17 reports the PnL and tail loss behind these claims.

A regime label applies only to the windows assigned to it, and a single high-volatility window cannot characterise a regime. The map therefore describes the evaluation year rather than a fitted boundary. The study's inference rests on the paired intervals of the primary statistics and not on any single regime cell.

VIII.B What the hedge buys and costs

The hedge is insurance. The study compares policies at three points along one trade-off between protection and cost, though not as a controlled hedge ablation. At full delta match on the opening band the tail is nearly flat and the maximum drawdown is a fraction of the unhedged one. The bill is an average negative hedge flow and a total PnL below the unhedged rule. At the learned seed's partial ratios the middle of that trade-off appears. The one-shot unhedged rules sit at the other end with a large tail and no premium, and the two rebalancing rules trade at similar tails. Across the compared policies the ordering is monotone in protection and in cost. Funding was positive on average at all measured points because the short collected it, but the mark loss on a short in a rising market and the perp fees outweighed that income. Table 13 reports the corresponding PnL, Sharpe ratio and tail loss.

The hedge does not improve the mean, it reshapes the tail, so its value depends on the risk budget against which the LP is managed. A mandate that treats tail risk as important as Sharpe will buy the protection that a mandate maximising return will refuse. The pre-registered rule gives tail risk equal standing, which is why a policy that fails the return clause and passes the tail clause is a partial success

rather than a failed idea.

VIII.C Why timing collapsed to a deployment decision

None of the trained seeds learned to time. They deployed one band and held it. The rebalancing benchmarks explain why. On this fee tier and in this year, frequent recentring repeatedly realises impermanent loss, and the fees collected do not cover it. Both active rules finish further behind the basket than any one-shot rule, despite trading far more (Table 12). Timing is not irrelevant in principle, but it is dominated in this market.

The study's configuration restricted the timing head after the early runs showed the collapse, so the estimator was never asked to learn a decision that the evidence indicates it cannot afford. That restriction is itself a finding about the pool, and it bounds the claim. A higher fee tier, cheaper execution or a market where price moves arrive more slowly could move the boundary, and the width choice substitutes for timing under those conditions. The action interface still admits the wider timing grid, so a future run can test the shift without a redesign.

The evidence base has limits that bound the claims. The failure modes were repaired in sequence rather than varied one at a time, so the ablation table reports the sequence and three targeted probes, not the marginal effect of any single design choice. The boundary is drawn from the realised 2024 windows on one pool pair. The 0.30 % tier is measured for the rule-based ladder but not retrained with the learned policy, and the regime with a single window is a case study. A factorial ablation of the repairs remains open, as do a controlled volatility and volume sweep and a learned policy on the contrast tier. All three are re-runs of the same testbed.

VIII.D Threats to validity

The simulator replays recorded events, so prices, liquidity and fee growth are real rather than modelled. Its assumption that the agent is marginal is the main simplification. The position never moves the price, and its fee share is proportional to active liquidity. That is the standard replay assumption and it is recorded alongside the results, but it cannot capture queueing at the tick level or the impact of an unusually large position. The gas model charges representative per-action units rather than transaction receipts, so gas is a defensible floor rather than an exact quote. The hedge uses a perpetual proxy whose funding and basis are representative rather than exact for a specific venue.

Table 18: Threats to validity and the scope it imposes.

Threat	Scope
One asset pair at two fee tiers	the boundary is measured within that scope and does not automatically transfer to other pairs
CVaR at 5 % from few episodes	reported with bootstrap intervals and the worst window, not treated as decisive alone
Regime labels from fixed thresholds	a labelling device for reporting and training, not a discovered state
Boundary sweep over several axes	inference rests on paired intervals on the pre-registered primary metric; secondary wins are labelled as such
Simulator and backtester both replay recorded states	claims hold under the stated cost model, not for live execution

Table 18 states the threats that bound what the study can claim and the scope it imposes. The parity check bounds the distance between the simulator and the backtester, not the distance between the backtester and reality. Where the data allow, the external part of the gap is quantified by reconciling fees and collects. Where they do not, latency, the privacy of order flow and MEV are discussed rather than measured. We therefore claim results about performance under risk on recorded market states under a stated cost model. We do not claim deployable profitability, optimality of the action grid, or that the policy would survive market conditions absent from 2022 to 2024.

IX Conclusion

Concentrated liquidity promises more fee income while the price stays inside a chosen band, and the same position that earns those fees is the position that loses to informed order flow. An LP who wants to do better must decide when to recentre, how wide to stay and how much risk to hedge, and those decisions are costly, uncertain and dependent on one another.

This study asks whether a risk-sensitive learned policy can make those decisions jointly better than the best hand-written rules, and where active management stops paying. Answering that required costs to be measured rather than assumed, so we built a testbed from three years of recorded trades, with an exact replay engine for ground truth, a fast simulator for training and a parity harness between them. The questions were where fees fail, why the first policies failed, whether one policy does better, and where active management pays.

The measured verdict is narrower than the bargain promised. On the 0.05 % pool in 2024 the liquidity leg of the rule ladder finished below the pure buy-and-hold basket, and the rebalancing rules lost the most. The learned policy deployed one band and held it without learning to time, failed the

Sharpe clause of the pre-registered rule against the frozen comparator (difference -0.733, interval $[-1.355, -0.041]$), and passed the tail clause against the unhedged rules. Its best seed matched the frozen comparator on Sharpe and led it on CVaR, while the delta-matched hedge bought a nearly flat tail at a carry cost. A higher fee tier turned the wide bands net positive while the rebalancing rules stayed behind, so the sign of passive provision is set by the fee tier rather than by the choice of rule. Learning did not make active management pay on this market. What the policy bought was defence, and the delta-matched hedge put a price on it.

What remains open follows from the boundary the windows draw. The action space can admit continuous width and an explicit exit decision, the learned policy can be trained on the $0.30\,\%$ fee tier, and the repairs can be probed one at a time in a factorial ablation. Latency, private order flow and the exact funding of a real venue can be modelled rather than proxied. Those runs share the testbed, and the remaining question is whether the learned policy pays in a market that charges more per swap or moves more slowly.

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Table 19: Notation. Subscript t marks a quantity at decision or event time t; the pool's discrete event index advances one row at a time through the recorded tape.

Symbol	Meaning
P_t	pool price at time t, in USDC per WETH
P_t^{ref}	reference exchange price used for the as-of join
s_t, $\sqrt{p}_t$	square-root price, the pool's native coordinate
i_t	tick index, monotone in the price; the engine maps between the two with the pool's decimals
L	liquidity of a position, in the contract's integer units
x_t, y_t	position amounts of USDC, the numeraire (token0), and WETH, the risky asset (token1)
$[i_\ell, i_u]$	lower and upper tick of the live band
c_t	centre of the band, in tick-spacing units from the current tick
w_t	half-width of the band, in tick-spacing units
κ_t	indicator that the agent acts (recentres) at time t
h_t	hedge ratio: short notional as a fraction of the LP's WETH delta
W_0	initial capital, 100 000 USDC per episode
V_t	mark value of the liquidity position
H_t	value of the opening deposit held as tokens, the HODL numeraire
G_t	cumulative gain, $V_t + \text{fees}_t - H_t$ before costs
f_t	fee income earned over step t, in USDC
ι	fee tier, in pips; 0.05 % is $\iota = 500$
σ	annualised realised volatility of the reference price
λ	weight on the risk penalty in the reward
α	CVaR level; the study reports $\alpha = 0.95$
CVaR_α	conditional value-at-risk at level α
γ	discount factor
g_t	gas paid at time t, in USDC
S_t	slippage paid at time t, in USDC
π_t	mark PnL of the short perpetual position
ϕ_t	funding paid or received on the short
τ_t	taker cost of adjusting the short
Δ	difference operator, $\Delta X_t = X_t - X_{t-1}$
δ	half-width of a band in log-price (Appendix C)
κ	concentration factor of a band relative to the full range
LVR	loss-versus-rebalancing rate of the position
$\mathcal{T}$	one decision interval, ten minutes in the study

A Notation and Glossary

Table 19 collects the symbols used in the derivations and the results. All monetary quantities are in USDC, the pool's token0, unless a caption says otherwise; prices are quoted in USDC per WETH, so a rise in P_t is an appreciation of the risky asset.

The report uses the following abbreviations. An automated market maker (AMM) quotes prices from a deterministic rule instead of an order book; Uniswap v3 is the concentrated-liquidity AMM studied here.[1] A liquidity provider (LP) supplies the pool. Loss-versus-rebalancing (LVR) is the arbitrage loss the LP bears when the pool price is stale relative to the wider market;[2] impermanent loss (IL) is the same

Table 20: The two pools and the recorded window. Deployment blocks are the factory creation blocks; the tape begins at the first event inside the window.

Setting	Primary pool	Contrast pool
Contract	`0x88e6a0…f5640`	`0x8ad599…e6d8`
Fee tier	0.05 % (500 pips)	0.30 % (3000 pips)
Tick spacing	10	60
Deployment block	12 376 729	12 370 624
Recorded window	2022-01-01 to 2024-12-31	2022-01-01 to 2024-12-31
Training window	2022-01-01 to 2023-12-31	for selection only
Evaluation window	2024-01-01 to 2024-12-31	2024-01-01 to 2024-12-31

gap measured against a passive holding. A Markov decision process (MDP) is the sequential-decision model. Proximal policy optimisation (PPO) is the training algorithm[32] and generalised advantage estimation (GAE) supplies its advantage estimates.[33] Conditional value-at-risk (CVaR) at level α is the mean of the worst $1-\alpha$ fraction of outcomes; value-at-risk (VaR) is the corresponding quantile.[9] Total value locked (TVL) is the value of assets in a pool, maximal extractable value (MEV) is the profit available to block builders from ordering transactions, and a perpetual swap (perp) is the derivative used for the hedge.

B Data Collection and Validation

The study replays recorded history rather than simulating a market from a fitted process, so the tape's provenance and its known defects are part of the result. This appendix records the pools, the feeds, the checks that were run and the two warnings that survived them.

B.A Pools and window

Table 20 lists the two pools. Both are USDC/WETH pools on Ethereum mainnet; the token ordering is fixed by address, so USDC is token0 and WETH is token1 in both, and all monetary quantities in the study are in USDC. The window is January 2022 to December 2024, with 2024 held out. The primary pool's higher activity is why it carries the learned policy; the contrast pool exists to price the fee tier.

B.B The recorded feeds

Each pool contributes an event tape of swaps, mints and burns; the window contains no collects whose position was opened inside it, so the tape's collect stream is empty and fee income is instead taken from the pool's fee-growth accounting, the settlement path the engine replays. Five global feeds join

Table 21: Recorded rows per stream for the primary and contrast pools over 2022–2024. The gas stream is chain-wide and shared; the reference and regime streams carry two symbols each, so the per-symbol count is half the row count; the lattice row counts boundary snapshot files, not rows.

Stream	Primary	Contrast	Note
Swaps	6 862 602	447 714	the priced event stream
Mints	119 781	53 510	liquidity added
Burns	130 516	53 630	liquidity removed
Collects	0	0	none inside the window
Gas	7 609 726	shared	base fee and priority fee per block
Reference	3 156 480	shared	one-minute bars, two symbols
Regime	3 156 480	shared	labels derived from the reference
Funding	3 288	shared	eight-hour perpetual rate
Lattice snapshots	2	2	boundary states per pool
Tape rows (swaps + mints + burns)	7 112 899	554 854	validation totals

the tape: one-minute reference prices, per-block gas prices, regime labels, boundary snapshots of the tick lattice, and eight-hour perpetual funding. Table 21 counts the rows for both pools. The reference and regime feeds carry two symbols, `ETHUSDC` and `ETHUSDT`; the loader keeps one per dataset (the deeper `ETHUSDT` feed) so no observation mixes the two series.

The training split of the primary pool holds roughly 4.86 million tape events, which is why the exact engine cannot train the policy and the fast simulator exists.

B.C Validation and the two warnings

Fourteen checks ran on each dataset: block ordering, key uniqueness, monotonic timestamps, absence of future data, block contiguity, swap-sign convention, tick and price consistency, tick spacing, liquidity conservation, reference coverage, regime completeness, fee reconciliation, external cross-checks and route cross-checks. Ten were conclusive, of which two returned warnings; four could not be completed because the tape holds no position lifecycle and no external magnitude feed.

The first warning concerns the sign convention. Of the primary pool's swaps, 128 of 6 862 602 report non-positive or same-signed token amounts, and 26 of 447 714 do so on the contrast pool. These are subgraph artefacts around zero-amount swaps; the replay treats a same-signed row as a transfer of the reported magnitudes and the affected rows are a fraction of a basis point of the tape.

The second warning concerns the edges of the reference feed. The first close lands at 59.999 seconds past the window's first minute and the last inside the window at the final minute's close, so an as-of join against the exact boundary leaves the opening and closing minute without a fresh bar. The loader

enforces the split wall and the replay warm-up covers the left edge, so no decision inside the window is affected.

Two further limitations matter for the reported results. First, the subgraph does not expose the pool's active liquidity on swap rows, so the replay reads the boundary snapshot and the fee-growth stream for liquidity state instead of the swap rows. Second, no completed position lifecycle sits inside the window, so the fee reconciliation against collects was skipped; fee income is instead taken from the protocol's own fee-growth accounting, which is the settlement path the engine replays exactly.

B.D Splits and evaluation windows

Training uses windows of thirty days resampled from the 2022–2023 lane, with the final 180 days of that lane carved out for checkpoint selection. The evaluation runs from 2024-01-01 to 2024-12-31 at midnight UTC, which tiles into twelve full thirty-day windows and a final partial window of five days, for thirteen episodes. Table 45 in Appendix F lists the realised market state of each evaluation window; the window boundaries are fixed before any policy is scored, so no comparison selects favourable periods after the fact.

C Derivations

This appendix carries the mathematics behind the study: the value of a concentrated position, the two loss measures, the concentration factor, the fee and hedge accounting, the risk measure and the policy update. The results in the main body use these expressions as definitions, so the appendix doubles as a statement of exactly what was measured.

C.A The value of a concentrated position

A constant-product pool with reserves (x, y) quotes the invariant $xy = k$ and a price $P = y/x$. A Uniswap v3 position replaces the full-range reserves with a liquidity constant L attached to a band $[p_\ell, p_u]$ in price space; the band's boundaries are tick indices i_ℓ, i_u, which correspond monotonically to p_ℓ, p_u through the pool's conversion between ticks and prices. The position's token amounts follow from the invariant in square-root coordinates. Writing $\sqrt{P}$ for the current square-root price,

$$x(P) = L\big(\sqrt{P} - \sqrt{p_\ell}\big), \tag{9}$$

$$y(P) = L\Big(\frac{1}{\sqrt{P}} - \frac{1}{\sqrt{p_u}}\Big), \tag{10}$$

so x is the USDC amount and y the WETH amount, matching the main body's token0/token1 convention. While $p_\ell \le P \le p_u$; outside the band the position holds only the token the price has moved toward. The mark value in USDC is $V(P) = x(P) + y(P)P$, which in range becomes

$$V(P) = L\Big(2\sqrt{P} - \sqrt{p_\ell} - \frac{P}{\sqrt{p_u}}\Big). \tag{11}$$

The deposit is fixed at W_0 and the position is opened at the entry price P_0 with the current tick as the band's centre, so the liquidity is pinned by $V(P_0) = W_0$:

$$L = W_0 \Big/ \Big(2\sqrt{P_0} - \sqrt{p_\ell} - \frac{P_0}{\sqrt{p_u}}\Big). \tag{12}$$

The position's exposure is the derivative of the value with respect to the price. In range,

$$\frac{\partial V}{\partial P} = y(P) = L\Big(\frac{1}{\sqrt{P}} - \frac{1}{\sqrt{p_u}}\Big), \qquad \frac{\partial^2 V}{\partial P^2} = -\frac{L}{2}\,P^{-3/2} < 0. \tag{13}$$

The LP is long the risky token while the price is low in the band, and short it while the price is high; the value is concave in the price because the position sells the asset that rises and buys the one that falls. Concavity is the source of the loss to arbitrage: no matter which way the price moves, the position underperforms a portfolio that would have rebalanced to the same exposure continuously.

C.B Impermanent loss

The HODL benchmark holds the entry amounts $(x_0, y_0) = (x(P_0), y(P_0))$ fixed, so its value is $H(P) = x_0 + y_0 P$. Impermanent loss is the gap

$$\mathrm{IL}(P) = V(P) - H(P) \;\le\; 0 \quad \text{for } P \neq P_0, \tag{14}$$

with equality at entry. Substituting (11) and the entry amounts gives the closed form used by the evaluation:

$$\frac{V(P)}{W_0} = \frac{2\sqrt{P} - \sqrt{p_\ell} - P/\sqrt{p_u}}{2\sqrt{P_0} - \sqrt{p_\ell} - P_0/\sqrt{p_u}}, \qquad \frac{H(P)}{W_0} = \frac{x_0}{W_0} + \frac{y_0}{W_0} P. \tag{15}$$

For a band symmetric in ticks around P_0, write $p_\ell = P_0/r$ and $p_u = P_0 r$ with $r = e^{\delta}$, where δ is the half-width in log-price and equals the half-width in ticks multiplied by the tick size. At the entry price the two legs are equal in value,

$$\frac{x_0}{W_0} = \frac{y_0 P_0}{W_0} = \frac{1}{2}, \qquad \frac{y_0}{W_0} = \frac{1}{2P_0}, \tag{16}$$

so the deposit is balanced at any symmetric width and $x_0/W_0 + y_0 P_0/W_0 = 1$ by construction.

C.C Concentration and loss-versus-rebalancing

Consider a price that follows a geometric Brownian motion, $\mathrm{d}P_t = \mu P_t \,\mathrm{d}t + \sigma P_t \,\mathrm{d}W_t$. Let R_t be the value of the continuously rebalanced portfolio that starts at $V(P_0)$ and holds the AMM's instantaneous exposure at every instant, $\mathrm{d}R_t = (\partial V/\partial P)\,\mathrm{d}P_t$. Itô's lemma gives

$$\mathrm{d}V_t = \frac{\partial V}{\partial P}\,\mathrm{d}P_t + \frac{1}{2}\frac{\partial^2 V}{\partial P^2}\,\sigma^2 P_t^2\,\mathrm{d}t, \tag{17}$$

so the gap to the rebalancing portfolio accumulates at

$$\mathrm{dLVR}_t = -\frac{1}{2}\frac{\partial^2 V}{\partial P^2}\,\sigma^2 P_t^2\,\mathrm{d}t \;\geq\; 0, \tag{18}$$

which is the loss-versus-rebalancing rate of Milionis et al.[2] Substituting the curvature of (13), an in-range concentrated position loses at the instantaneous rate

$$\mathrm{dLVR}_t = \frac{\sigma^2}{4}\,L\,\sqrt{P_t}\,\mathrm{d}t. \tag{19}$$

The full-range position is the special case $p_\ell \to 0$, $p_u \to \infty$ with $V = 2L\sqrt{P}$, for which (19) collapses to $\sigma^2 V/8$, the rate in the LVR literature. Dividing the two rates and evaluating at the entry price, with both positions pinned to the same capital W_0, gives the position's concentration factor,

$$\kappa = \frac{\mathrm{dLVR}^{\text{concentrated}}}{\mathrm{dLVR}^{\text{full range}}} = \frac{1}{1-(p_\ell/p_u)^{1/4}} = \frac{1}{1-e^{-\delta/2}}, \tag{20}$$

so a band of half-width $\delta = 0.012$ in log-price, the default open band of the replay (half-width 120 ticks), carries an instantaneous LVR rate about 167 times the full-range rate while it is in range. The factor is why a narrow band earns more fees and loses more to arbitrage at the same time, and why the fee tier, not the width alone, decides whether the position clears its own LVR.

Fees modify the arbitrage relation. With a proportional fee ι, a riskless arbitrageur only trades when the pool price deviates from the external price by more than the fee, so the pool price wanders inside a band of width proportional to ι around the external price before arbitrage is profitable. Fee income and the residual LVR are then two sides of the same threshold, which is the object the empirical fee-to-LVR ratio in Appendix F measures; the fee-adjusted arbitrage condition is derived by Milionis et al.[22]

C.D Fee income and the liquidity share

While the price is inside the band, the position earns a proportional share of every swap fee. If the position's liquidity is L and the pool's active liquidity is L_{active}, that share is the ratio L/L_{active} on each crossed tick segment. The exact engine does not approximate this ratio at the pool level: each swap is apportioned across the tick segments it crosses, and the position's fee accrual is the integer change in the inside fee-growth accumulator multiplied by its liquidity. Fee income over a step is therefore

$$f_t = \sum_{\text{segments } s} L_s \cdot \left(\Delta\text{feeGrowthInside}_s\right) / 2^{128}, \tag{21}$$

with $L_s = 0$ outside the band and the 2^{128} divisor the protocol's fixed-point normalisation. The *marginal-agent* assumption of the simulator replaces the sum with the position's share of the global fee growth while in range, which removes the segment bookkeeping at the cost of a small approximation bounded by the parity harness.

C.E The hedge leg

The hedge shorts $h_t\ y(P_t)$ units of WETH through a perpetual swap, where y is the LP's WETH amount. Its mark PnL over a step is

$$\pi_t = -\,h_{t-1}\,y(P_{t-1})\,\big(P_t - P_{t-1}\big), \tag{22}$$

funding is $\phi_t = +h_{t-1}y(P_{t-1})P_{t-1}\,\rho_t$ when the eight-hour rate ρ_t, scaled to the step, is positive (the short collects it), and adjusting the position costs a taker fee $\tau_t \le 0$ proportional to the change in short notional. The joint strategy value is $V_t + \text{fees}_t + \pi_t + \phi_t + \tau_t - g_t - S_t$. Because the hedge is linear in the price and the LP value is concave, the hedge removes the first-order exposure but cannot remove the curvature term (19): a delta-neutral LP still pays LVR. This is the structural reason the empirical hedge cuts the tail without repairing the mean.

C.F Risk measure and the training objective

For a loss $L = -G$ on the episode gain G, conditional value-at-risk at level α is the mean of the worst $1-\alpha$ tail,

$$\mathrm{CVaR}_\alpha(L) = \frac{1}{1-\alpha}\int_\alpha^1 \mathrm{VaR}_u(L)\,\mathrm{d}u = \min_{\nu\in\mathbb{R}}\left\{\nu + \frac{1}{1-\alpha}\mathbb{E}\big[(L-\nu)^+\big]\right\}, \tag{23}$$

the convex representation of Rockafellar and Uryasev;[9] CVaR is a coherent risk measure in the sense of Artzner et al.[30] The reward subtracts a multiple of a trailing empirical estimate of (23) on per-step joint equity changes, with a one-day window, so the policy's gradient sees the same tail statistic the evaluation reports.

Policy optimisation uses the clipped surrogate of PPO.[32] With probability ratio $r_t(\theta)$ between the new and old policies and advantage $\hat{A}_t$ from generalised advantage estimation,[33]

$$\mathcal{L}(\theta) = \mathbb{E}\Big[\min\big(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta),1-\epsilon,1+\epsilon)\hat{A}_t\big)\Big] - c_v\,\mathbb{E}\big[(V_\theta(s_t)-\hat{V}_t)^2\big] + c_e\,\mathbb{E}\big[H(\pi_\theta(\cdot\mid s_t))\big], \tag{24}$$

with clip $\epsilon = 0.15$, value coefficient c_v and entropy bonus c_e. The advantage is the exponentially weighted sum of one-step temporal-difference residuals with parameter $\lambda_{\mathrm{GAE}} = 0.95$. The full training settings are the subject of Appendix D.

D Testbed Specification

The report's claims rest on two engines and the check between them. This appendix specifies both engines, the parity harness, the cost models and the learning interface in enough detail to rebuild them.

D.A The exact replay engine

The exact engine consumes the recorded tape at its own event granularity and asks the policy for a decision every ten minutes. Each decision is executed at the price observed *before* the interval advances, matching the latency a live LP faces and keeping the simulator and the backtester on the same convention. The engine's state is the pool's integer tick lattice: liquidity is minted and burned on the exact ticks the tape reports, and a swap is apportioned across the tick segments it crosses. Fee accrual comes from the protocol's own fee-growth accumulators, which are 128-bit fixed-point integers that increase monotonically with every swap; the position's fee over a step is its liquidity multiplied by the change in the inside fee-growth value divided by the protocol's 2^{128} normalisation, exactly as the contract computes it. The ledger therefore satisfies an accounting identity by construction,

$$\text{fees} + \text{IL} - \text{gas} - \text{slippage} = \text{strategy excess over the opening basket}, \tag{25}$$

and the identity is checked at each step of a replay.

Three numerical details matter for exactness. First, liquidity arithmetic is done in Python integers, not floats, because a large gross liquidity in floating point carries a residue that can round a small delta to zero. Second, the initial position is deployed before the first decision for all policies, so the HODL benchmark has a definite token mix; the pure basket holds the same mix as tokens. Third, a rebalance that the position cannot afford is recorded as rejected and the position is held, mirroring an on-chain revert rather than forcing a sale.

D.B The fast simulator

Training needs roughly 10^6 decisions per seed, which the exact engine cannot supply within the study's budget. The simulator mirrors the same economics in floating point under a marginal-agent assumption: the agent's position never moves the pool price, and its fee share is its fraction of the active liquidity while in range. This removes the segment-level replay and raises throughput by more than an order

Table 22: Cost-model parameters. Gas units are per action and sweepable in the configuration; the tip and the slippage impact are the shipped values.

Parameter	Value	Charged on
Mint gas units	460 000	opening or rebalancing a position
Burn gas units	215 000	closing the position
Collect gas units	130 000	fee collection
Rebalance swap units	150 000	the swap inside a rebalance
Tip surcharge	3 % of the base fee	every action
Slippage impact	5 basis points	the swapped inventory
Hedge taker fee	5 basis points	short notional adjustments

of magnitude. The simulator runs either on the recorded events, the mode used for the results in this report, or on a regime-switching jump process calibrated to the training years; the calibrated mode is not exercised here because the study keeps the evaluation on recorded states.

D.C The parity harness

Two engines are only useful together if their disagreement is bounded. The harness feeds both the same prices, positions and gas series and compares fees, impermanent loss, PnL, gas and slippage per step against a tolerance of 10^{-3} USDC. On the recorded data the residual fee drift is on the order of 10^{-7} USDC per step per seed, recorded with the engine validation (8.9×10^{-8}, 1.4×10^{-7}, 1.0×10^{-7}, 1.1×10^{-7} for four seeds); impermanent loss matches exactly, and the pooled error is dominated by a single seed's tail but remains two orders of magnitude inside tolerance.

A separate suite of thirteen look-ahead probes attacks the composed system at the seams where a look-ahead error would be easiest to hide: the split wall for slice access and tape access, the reference-price wall, the as-of regime probe, the gas and observation builders against future mutation, the sentinel values, episode truncation and the backtester's own future mutation. All thirteen pass on the shipped code.

D.D Cost models

Gas is priced at each block's base fee plus a 3 % tip surcharge, converted at the block's reference ETH price, and charged as representative per-action units rather than per-transaction receipts. Table 22 lists the units. Slippage is proportional to the notional actually swapped: the pool fee tier plus a fixed five-basis-point impact. The hedge pays a five-basis-point taker fee on notional changes and the recorded eight-hour funding rate on the short.

Table 23: The observation vector, in order. The right column gives the feature in words; the regime slot is the only categorical entry.

#	Slot	Feature
1	price	pool price
2	sqrt price	square-root price
3	tick	current tick
4	r_{1h}	one-hour reference return
5	r_{24h}	twenty-four-hour reference return
6	σ_{24h}	twenty-four-hour realised volatility
7	in range	whether the live band covers the price
8	value	position value in USDC
9	fees0	uncollected fees in USDC
10	fees1	uncollected fees in WETH
11	IL	IL against the opening basket, as a fraction of capital
12	liquidity	active pool liquidity
13	fee tier	pool fee tier in pips
14	gas	recent base fee plus priority fee
15	ETH price	recent reference price
16	regime	as-of regime ordinal
17	step	decision index in the episode
18	remaining	decisions left in the episode
19	d_ℓ	distance to the lower bound, in tick spacings
20	d_u	distance to the upper bound, in tick spacings
21	width	band width, in tick spacings
22	hedge ratio	current short ratio
23	hedge equity	hedge account equity, as a fraction of capital
24	funding	recent eight-hour funding rate

Table 24: The restricted action grid of the study. Widths are half-widths in tick spacings; the hedge ratio is the short notional as a fraction of the LP's WETH delta, and it is chosen at each decision.

Head	Options	Meaning
LP action	hold, deploy 50, 250, 1000	deploy once, then hold
Hedge ratio	0, 0.25, 0.5, 0.75, 1.0	short share of the delta

D.E The learning interface

The observation is a fixed-order vector of twenty-four scaled features. Table 23 lists them in order. The heavy-tailed dimensions are log-compressed so all inputs reach the policy at order one. The regime slot is the ordinal of the as-of label; the ablation arm in Appendix F freezes that slot to a constant.

Actions are factored. The general grid pairs nine centre offsets with ten widths and a hold option; the study's configuration, after the timing finding, restricts the centre to zero and the width to three arms, and adds the hedge ratio as an independent head. An action is therefore one of four LP choices (hold or one of three widths) crossed with five hedge ratios, a MultiDiscrete$([4, 5])$ space. Table 24 lists the arms.

Table 25: Optimisation settings for the learned policies in the study.

Setting	Value
Algorithm	clipped PPO with GAE
Seeds	0 to 4, trained independently
Timesteps per seed	1 000 000
Discount factor γ	0.9999
GAE parameter λ_{GAE}	0.95
Clip range ϵ	0.15
Learning rate	10^{-4}
Batch size	512
Epochs per update	4
Entropy coefficient	0.001
Target KL	0.01
Network	two hidden layers of 256 units
Parallel environments	4, one process each
Checkpoint interval	100 000 steps
Selection metric	validation Sharpe, last 180 days of training
Risk penalty	CVaR at 5 %, weight $\lambda = 0.05$, one-day window

D.F Training settings

Table 25 lists the optimisation settings. Five seeds are trained per configuration and the checkpoint with the highest validation Sharpe on the last 180 days of the training lane is the one evaluated. The configuration was fixed before the evaluation year was scored.

E Complete Per-Window Results

The main body reports means over the evaluation windows. This appendix reports the full window history by rule, so the reader can see the dispersion behind the means. Each table gives the window number, its dominant regime, PnL in USDC, the annualised Sharpe ratio, maximum drawdown as a percentage of capital, CVaR at the 5 % level in USDC per tape event, and the liquidity leg's net excess over the opening basket. The windows are the same for all rules; the final window is the five-day partial period, which is why its magnitudes are smaller.

The tables cover the five benchmark rules, the five learned seeds of the main run, and the two scripted recentring rules that complete the pre-registered ladder. The learned seeds are evaluated at the checkpoint selected by validation Sharpe on the training lane, never on these windows.

Table 26: Per-window evaluation of hodl. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	+7 096	2.68	−7.7	−42.5	+5 697
2	bull	+2 608	2.23	−3.9	−19.1	−18 319
3	bull	+11 103	2.79	−11.9	−24.2	+8 532
4	bear	−3 400	−0.33	−17.8	−90.6	+623
5	bear	+900	0.46	−12.1	−59.8	−7 639
6	bull	−5 791	−1.64	−13.4	−62.2	−551
7	bear	+1 243	0.59	−16.0	−66.6	+2 812
8	high volatility	−23 000	−3.17	−34.0	−94.5	−10 937
9	bear	+6 128	1.64	−12.0	−76.6	+2 671
10	sideways	+2 201	0.78	−11.9	−87.4	+5 075
11	bull	+4 976	4.05	−4.4	−17.9	−14 342
12	bull	+8 777	3.30	−8.6	−35.9	+7 560
13	bear	−3 283	−4.89	−5.1	−98.1	−1 372

Table 27: Per-window evaluation of passive_narrow. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	+7 698	3.53	−6.9	−36.7	+6 299
2	bull	+3 495	3.88	−3.0	−15.7	−17 432
3	bull	+11 397	4.14	−8.2	−27.2	+8 826
4	bear	−3 819	−0.40	−17.5	−89.4	+204
5	bear	+3 511	1.18	−11.4	−57.1	−5 028
6	bull	−6 859	−2.02	−12.5	−60.9	−1 620
7	bear	+2 395	0.90	−15.6	−66.9	+3 964
8	high volatility	−22 542	−3.09	−33.5	−94.4	−10 479
9	bear	+7 592	2.09	−11.0	−71.6	+4 135
10	sideways	−582	0.05	−11.1	−81.9	+2 293
11	bull	+4 259	4.20	−3.9	−17.3	−15 059
12	bull	+8 346	3.57	−7.9	−34.7	+7 128
13	bear	−820	−1.17	−4.1	−86.5	+1 091

Table 28: Per-window evaluation of passive_wide. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	+4 578	2.41	−6.6	−39.7	+3 179
2	bull	+7 087	7.70	−2.4	−19.9	−13 841
3	bull	+9 082	3.81	−7.8	−46.7	+6 511
4	bear	−785	0.09	−15.2	−71.4	+3 239
5	bear	+8 390	3.27	−8.4	−42.4	−149
6	bull	−4 878	−1.98	−9.7	−45.8	+361
7	bear	+1 255	0.63	−12.8	−56.1	+2 824
8	high volatility	−19 123	−2.55	−31.2	−93.9	−7 060
9	bear	+5 656	2.00	−9.7	−57.5	+2 200
10	sideways	−1 193	−0.28	−9.0	−62.0	+1 682
11	bull	+6 906	6.07	−5.7	−23.9	−12 412
12	bull	+5 697	2.62	−9.6	−48.9	+4 479
13	bear	−1 573	−3.94	−3.3	−59.0	+338

Table 29: Per-window evaluation of full_range_v2. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	+1 657	0.89	−10.5	−48.7	+258
2	bull	+19 745	8.87	−4.2	−51.9	−1 182
3	bull	+2 958	1.19	−13.2	−65.0	+387
4	bear	−3 848	−1.17	−12.3	−51.1	+175
5	bear	+8 563	3.21	−6.8	−45.1	+24
6	bull	−5 139	−2.81	−8.5	−36.2	+100
7	bear	−1 338	−0.45	−10.3	−45.8	+231
8	high volatility	−12 642	−3.39	−20.5	−54.5	−579
9	bear	+3 747	1.48	−8.8	−53.9	+291
10	sideways	−2 756	−1.16	−7.8	−50.6	+118
11	bull	+17 712	6.57	−6.8	−73.2	−1 606
12	bull	+1 618	0.79	−12.9	−65.2	+400
13	bear	−1 903	−5.72	−3.0	−50.3	+8

Table 30: Per-window evaluation of hedged_hodl. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	−2 237	−2.25	−3.8	−4.1	+5 697
2	bull	−594	−4.88	−1.0	−1.5	−18 319
3	bull	−3 071	−0.60	−7.5	−8.2	+8 532
4	bear	−3 260	−9.44	−3.7	−3.1	+623
5	bear	−2 405	−4.86	−2.4	−0.6	−7 639
6	bull	−1 590	−7.25	−2.1	−2.3	−551
7	bear	−926	−4.30	−1.3	−2.0	+2 812
8	high volatility	−1 330	−9.03	−1.5	−1.4	−10 937
9	bear	−1 974	−6.65	−2.3	−3.9	+2 671
10	sideways	−4 054	−6.73	−4.5	−4.7	+5 075
11	bull	−1 479	−4.25	−2.2	−4.2	−14 342
12	bull	−3 057	−6.39	−3.2	−4.3	+7 560
13	bear	+30	4.66	−0.1	−0.5	−1 372

Table 31: Per-window evaluation of seed 0. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	+1 370	1.35	−4.6	−11.9	+6 299
2	bull	+1 613	3.45	−1.5	−8.3	−17 432
3	bull	+2 832	1.55	−5.7	−13.1	+8 826
4	bear	−3 635	−1.39	−9.3	−36.8	+204
5	bear	−1 554	−0.78	−5.8	−28.8	−5 028
6	bull	−1 400	−0.99	−5.0	−22.1	−1 620
7	bear	+276	0.29	−8.0	−28.3	+3 964
8	high volatility	−9 133	−2.99	−17.1	−41.2	−10 479
9	bear	−1 277	−0.59	−6.5	−34.0	+4 135
10	sideways	−131	0.07	−5.7	−36.4	+2 293
11	bull	+1 637	3.13	−2.1	−9.6	−15 059
12	bull	−95	−0.08	−2.0	−6.6	+7 128
13	bear	−684	−2.80	−2.1	−34.8	+1 091

Table 32: Per-window evaluation of seed 1. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	+488	0.53	−5.1	−21.6	+2 591
2	bull	+3 679	6.26	−1.8	−12.4	−12 085
3	bull	+5 050	3.50	−4.4	−28.1	+5 280
4	bear	−1 317	−0.55	−7.0	−34.6	+2 690
5	bear	+3 924	2.67	−6.1	−23.2	−98
6	bull	−2 924	−2.54	−5.2	−22.5	+309
7	bear	−3 021	−1.73	−9.1	−29.4	+2 308
8	high volatility	−2 981	−0.64	−13.5	−51.3	−5 162
9	bear	+4 537	3.01	−4.4	−29.9	+1 816
10	sideways	−3 697	−2.34	−6.6	−34.4	+1 367
11	bull	+3 383	5.35	−3.1	−13.7	−10 181
12	bull	+2 919	2.44	−5.5	−26.9	+3 658
13	bear	−1 868	−5.25	−3.5	−52.2	+1 091

Table 33: Per-window evaluation of seed 2. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	+2 714	1.53	−6.5	−26.2	+34
2	bull	+4 173	3.93	−2.9	−21.9	−10 055
3	bull	−613	−0.25	−5.9	−20.6	−1 472
4	bear	−4 369	−1.45	−11.1	−41.4	+2 582
5	bear	−4 105	−1.56	−9.2	−37.0	−8 752
6	bull	−1 744	−1.10	−6.6	−23.2	+2 460
7	bear	−2 726	−0.99	−12.1	−38.1	−2 257
8	high volatility	−14 186	−3.16	−25.8	−50.8	−7 602
9	bear	+2 218	1.00	−9.2	−41.5	+3 329
10	sideways	−3 599	−1.57	−9.0	−46.2	−1 479
11	bull	+4 708	3.68	−3.3	−28.5	−9 005
12	bull	+1 624	1.43	−3.9	−21.8	−1 036
13	bear	−2 176	−5.64	−3.8	−55.6	−1 471

Table 34: Per-window evaluation of seed 3. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	+1 280	2.07	−2.3	−13.3	+2 591
2	bull	+2 446	8.21	−0.6	−6.0	−12 085
3	bull	+2 719	3.13	−2.6	−16.9	+5 280
4	bear	−810	−0.52	−4.1	−21.3	+2 690
5	bear	+1 164	1.68	−2.4	−11.9	−98
6	bull	−2 135	−2.77	−3.9	−14.2	+309
7	bear	+207	0.33	−3.7	−16.8	+2 308
8	high volatility	−5 969	−3.40	−7.3	−27.8	−5 162
9	bear	+1 708	2.07	−2.6	−16.9	+1 816
10	sideways	−1 132	−1.25	−3.9	−18.4	+1 367
11	bull	+1 996	6.53	−1.5	−7.3	−10 181
12	bull	+439	0.54	−5.2	−19.7	+3 658
13	bear	+363	1.89	−2.0	−30.8	+272

Table 35: Per-window evaluation of seed 4. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	+5 034	3.61	−5.1	−27.2	+2 591
2	bull	+5 832	8.03	−1.8	−16.5	−12 085
3	bull	+3 168	2.40	−4.4	−26.3	+5 280
4	bear	−5 429	−1.90	−11.1	−42.1	+2 690
5	bear	−3 000	−1.66	−6.1	−27.8	−98
6	bull	−2 904	−2.14	−5.7	−24.8	+309
7	bear	+2 397	1.37	−9.2	−35.5	+2 308
8	high volatility	−17 926	−3.84	−22.9	−61.2	−5 162
9	bear	+1 675	0.95	−7.3	−37.4	+1 816
10	sideways	−2 725	−1.41	−6.6	−41.1	+1 367
11	bull	+5 996	6.71	−4.5	−19.6	−10 181
12	bull	+6 859	4.63	−3.4	−32.9	+3 658
13	bear	−1 335	−5.45	−2.4	−36.8	+272

Table 36: Per-window evaluation of `tau_reset`. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	−12 833	−4.42	−16.9	−52.1	−14 232
2	bull	+5 517	2.07	−12.0	−59.2	−15 411
3	bull	−40 086	−11.25	−44.8	−63.0	−42 657
4	bear	−32 178	−9.56	−35.5	−47.8	−28 155
5	bear	−11 317	−3.44	−12.2	−46.6	−19 855
6	bull	−12 663	−5.98	−15.1	−38.6	−7 424
7	bear	−13 808	−5.18	−17.5	−47.5	−12 239
8	high volatility	−38 321	−7.33	−40.4	−61.6	−26 259
9	bear	−11 587	−3.66	−16.1	−58.4	−15 043
10	sideways	−15 085	−5.96	−17.0	−57.0	−12 210
11	bull	−17 667	−6.16	−22.0	−72.9	−36 985
12	bull	−27 508	−8.95	−30.8	−64.9	−28 725
13	bear	−2 305	−5.70	−4.4	−59.8	−394

Table 37: Per-window evaluation of `cost_aware`. PnL, CVaR and net are in USDC; drawdown is a percentage of capital; net is the liquidity leg's excess over the opening basket.

Window	Regime	PnL	Sharpe	maxDD (%)	CVaR	Net
1	sideways	−692	0.05	−12.9	−64.6	−2 090
2	bull	+21 818	9.86	−4.1	−40.1	+891
3	bull	+285	0.40	−14.8	−78.2	−2 286
4	bear	−9 356	−2.01	−21.0	−55.4	−5 333
5	bear	−1 566	−0.21	−11.5	−50.6	−10 105
6	bull	−7 271	−2.83	−11.3	−44.0	−2 032
7	bear	−346	0.19	−18.2	−74.2	+1 223
8	high volatility	−24 221	−4.07	−31.9	−60.4	−12 158
9	bear	+6 430	1.94	−10.5	−65.2	+2 974
10	sideways	−7 369	−1.83	−11.6	−68.0	−4 494
11	bull	+6 044	2.42	−10.5	−63.2	−13 274
12	bull	−7 231	−1.46	−18.8	−80.7	−8 449
13	bear	−2 419	−5.57	−5.1	−59.4	−508

Table 38: Static one-shot widths on the training split, mean over the twenty-five training windows. The width with the best training Sharpe is frozen as the comparator.

Policy	PnL	Sharpe	CVaR	Fees	IL	Net
static 50	−1 486	+2.197	−48.01	4 759	−6 229	−1 529
static 250	−1 147	+1.448	−48.73	2 542	−3 673	−1 190
static 1000	−210	+0.804	−48.55	1 049	−1 243	−253

Table 39: The same static widths on the thirteen evaluation windows, mean. Width 50 is the frozen comparator chosen on the training split.

Policy	PnL	Sharpe	CVaR	Fees	IL	Net
static 50	+1 082	+1.296	−56.95	4 266	−5 435	−1 206
static 250	+1 732	+1.591	−51.03	2 637	−3 156	−557
static 1000	+2 147	+0.942	−52.41	919	−1 023	−142

F Ablation and Boundary Tables

The design section lists the probes that complete the study; this appendix carries their full tables. All tables are generated from the evaluation records by the reproduction program, so the numbers can be checked line by line against the analysis artefacts.

F.A Static width selection

The protocol freezes the static width with the best Sharpe ratio on the training split. Table 38 reports the three widths on the training windows and Table 39 their evaluation counterparts. The width of 50 tick spacings wins the training selection, and its evaluation row is the frozen comparator used in the main text; the width of 250 would have scored better on the evaluation year, which is exactly the selection risk the pre-registered rule guards against.

F.B The scripted recentring rules

The ladder's two active rules complete the comparison. Both recentre the band when the price moves; one does so whenever the price leaves the live band, the other at most daily and only when the estimated fee gain covers gas. Table 40 reports their means, and their per-window rows are in Appendix E. They are the empirical case against timing on this pool: high fee income, larger realised losses.

Table 40: The two scripted recentring rules completed for the study, mean over the thirteen evaluation windows, with the same metrics as the main ladder.

Policy	PnL	Sharpe	CVaR	Rebalances	Fees	Net
tau_reset	−17 680	−5.810	−56.10	122.85	43 029	−19 968
cost_aware	−1 992	−0.241	−61.84	22.00	19 833	−4 280

Table 41: Friction ablation on the primary pool's one-shot wide band, mean over the thirteen evaluation windows. The variant switches one cost term off or re-prices gas at a flat 30 gwei base fee plus a 3 % tip.

Variant	PnL	Sharpe	Fee income	IL	Gas	Slippage
full	+1 623	+1.526	2 921	−3 549	−36	−1
no gas	+1 659	+1.526	2 921	−3 549	+0	−1
no slippage	+1 624	+1.526	2 921	−3 549	−36	+0
no IL	+5 172	+1.647	2 921	+0	−36	−1
flat gas	+1 583	+1.526	2 921	−3 549	−76	−1

F.C Friction ablation

The friction ablation removes one cost term at a time from the same backtest ledger, so the differences are exact. Table 41 runs it on a one-shot band, Table 42 on the recentring rule, and Table 43 repeats the first on the higher fee tier. Removing impermanent loss moves the one-shot band by thousands of USDC and the recentring rule by tens of thousands, while gas and slippage together move the one-shot band by less than forty. The cost that decides the result is the loss to arbitrage, not execution.

F.D Fee-tier contrast

Table 44 repeats the whole rule ladder on the 0.30 % pool. The fee tier is the one axis that flips the sign of the passive arithmetic: the wide, full-range and static bands finish ahead of the opening basket, while both recentring rules still lose several thousand USDC. The contrast pool's tape is smaller and its regime path is not identical to the primary pool's, so the comparison is between fee tiers on the same recorded year, not a controlled experiment on one tape.

Table 42: Friction ablation on the primary pool's recentring rule, mean over the thirteen evaluation windows. The variant switches one cost term off or re-prices gas at a flat 30 gwei base fee plus a 3 % tip.

Variant	PnL	Sharpe	Fee income	IL	Gas	Slippage
full	−17 680	−5.810	43 029	−48 904	−8 772	−5 321
no gas	−8 908	−3.016	43 029	−48 904	+0	−5 321
no slippage	−12 359	−3.908	43 029	−48 904	−8 772	+0
no IL	+31 224	+12.841	43 029	+0	−8 772	−5 321
flat gas	−18 303	−6.318	43 029	−48 904	−9 395	−5 321

Table 43: Friction ablation on the contrast pool's one-shot wide band, mean over the thirteen evaluation windows. The variant switches one cost term off or re-prices gas at a flat 30 gwei base fee plus a 3 % tip.

Variant	PnL	Sharpe	Fee income	IL	Gas	Slippage
full	+1 808	+3.328	3 180	−2 787	−35	−20
no gas	+1 843	+3.328	3 180	−2 787	+0	−20
no slippage	+1 828	+3.328	3 180	−2 787	−35	+0
no IL	+4 595	+3.535	3 180	+0	−35	−20
flat gas	+1 767	+3.328	3 180	−2 787	−76	−20

Table 44: The rule ladder on the 0.30 % contrast pool, mean over the thirteen evaluation windows. The static widths are the price-equivalent of the primary pool's ladder at this pool's spacing of 60.

Policy	PnL	Sharpe	CVaR	Fees	IL	Net
hodl	+1 361	+2.062	−239.7	5 341	−5 450	−109
passive_narrow	+1 432	+2.886	−224.6	4 663	−4 651	−38
passive_wide	+1 808	+3.328	−190.4	3 180	−2 787	+338
full_range_v2	+2 188	+1.297	−188.1	403	+372	+719
static 8	+1 432	+2.886	−224.6	4 663	−4 651	−38
static 42	+1 893	+3.448	−186.8	2 823	−2 344	+423
static 167	+2 204	+1.965	−185.7	1 015	−225	+734
tau_reset	−20 702	−15.894	−218.3	36 924	−39 767	−22 172
cost_aware	−2 101	−0.265	−280.0	14 900	−15 883	−3 571

F.E Boundary sweep

The sweep reruns the ladder at a quarter and four times the recorded gas cost and at a tenth and ten times the capital. Table 45 first records the market state of each window: annualised realised volatility, swap notional, and the analytical full-range LVR floor. The four level tables then report the reruns. Two findings stand out. First, no gas price or capital level turns either recentring rule positive; cheap execution alone does not revive timing, and a larger book scales fees and losses together. Second, the best validation seed's (seed 3) Sharpe ratio stays between 1.40 and 1.46 at each level, so its risk-adjusted result is not an artefact of the tested cost range.

F.F Regime-feature ablation

The final probe retrains the same recipe across the five seeds with the regime slot of the observation frozen to a constant. Table 50 reports the arm. Its mean Sharpe is 0.837 against 0.563 with the live feature, a paired difference of +0.274 whose interval $[-0.13, +0.68]$ includes zero, while its mean CVaR is −32.89 USDC against −27.63 USDC, a difference of −5.26 USDC with interval $[-6.21, -4.22]$ that excludes zero. The live label therefore buys tail protection without a detectable return cost. The arm also diverges more across seeds: one seed never deploys, two widen to the 1000-spacing band, and the

Table 45: Recorded market state per evaluation window: annualised realised volatility from the reference series, swap notional, and the analytical loss-versus-rebalancing floor $0.125\,\sigma^2 Vt$ at 100 000 USDC for a full month; the final window is partial, so its fee comparison rescales the floor to the window length.

Window	Regime	σ	Volume (USDC)	LVR floor (USDC)
1	sideways	0.600	6 424 599 898	+370.2
2	bull	0.577	6 406 706 195	+341.9
3	bull	0.784	12 209 759 325	+631.7
4	bear	0.703	9 789 688 507	+507.7
5	bear	0.641	10 359 931 592	+421.6
6	bull	0.463	7 792 755 007	+220.0
7	bear	0.579	7 125 621 906	+344.1
8	high volatility	0.918	6 781 740 810	+865.5
9	bear	0.568	4 289 496 437	+330.9
10	sideways	0.490	3 990 759 064	+246.2
11	bull	0.717	9 323 871 454	+528.1
12	bull	0.735	9 428 225 648	+554.8
13	bear	0.546	1 041 805 188	+305.6

Table 46: Boundary sweep at gas 0.25x, mean over the thirteen evaluation windows. PnL and CVaR are in USDC.

Policy	PnL	Sharpe	CVaR	Rebalances
hodl	+735	+0.65	−59.7	0.00
passive_wide	+1 651	+1.53	−51.3	1.00
tau_reset	−11 613	−3.72	−57.7	122.85
cost_aware	−2 953	−0.54	−61.1	24.85
seed 3	+203	+1.42	−17.0	1.00

Table 47: Boundary sweep at gas 4x, mean over the thirteen evaluation windows. PnL and CVaR are in USDC.

Policy	PnL	Sharpe	CVaR	Rebalances
hodl	+735	+0.65	−59.7	0.00
passive_wide	+1 509	+1.53	−51.3	1.00
tau_reset	−40 032	−12.85	−50.0	117.54
cost_aware	−1 419	+0.01	−60.2	7.08
seed 3	+65	+1.42	−17.0	1.00

Table 48: Boundary sweep at capital 10k, mean over the thirteen evaluation windows. PnL and CVaR are in USDC.

Policy	PnL	Sharpe	CVaR	Rebalances
hodl	+81	+0.68	−6.0	0.00
passive_wide	+128	+1.53	−5.1	1.00
tau_reset	−6 764	−15.37	−4.2	89.38
cost_aware	−195	−0.10	−6.1	3.31
seed 3	−13	+1.46	−1.7	1.00

Table 49: Boundary sweep at capital 1m, mean over the thirteen evaluation windows. PnL and CVaR are in USDC.

Policy	PnL	Sharpe	CVaR	Rebalances
hodl	+1 384	+0.46	−595.0	0.00
passive_wide	+16 213	+1.51	−513.2	1.00
tau_reset	−162 687	−5.47	−559.4	122.85
cost_aware	−54 962	−1.35	−602.9	24.85
seed 3	+2 003	+1.40	−170.3	1.00

Table 50: The regime-feature ablation arm, mean over the thirteen evaluation windows. The regime input is frozen to a constant for training and evaluation.

Policy	PnL	Sharpe	CVaR	Rebalances
seed 0	−1 303	−0.059	−45.23	0.00
seed 1	+1 223	+0.528	−34.12	1.00
seed 2	−428	+0.583	−30.67	1.00
seed 3	+1 824	+2.356	−30.66	1.00
seed 4	+489	+0.777	−23.78	1.00

seed with the highest validation score finishes worst out of sample.

G Reproducibility

G.A Software

The project is written in Python 3.14 and managed with `uv`, which pins the environment from a lock file. The simulation and learning stack uses `numpy`, `polars` and `pyarrow` for data handling, `gymnasium` for the environment interface,[35] and `stable-baselines3` for the PPO trainer.[34] The report is built from LaTeX with a wrapper that selects among `tectonic`, `latexmk` and `pdflatex`.

G.B Program inventory

Table 51 lists the programs and their outputs. The analysis programs are deterministic given the artefacts they read; the learning runs are seeded. The repository exposes three console commands (data, simulator and dashboard), and the reproduction wrapper `scripts/report/reproduce.sh` is the single main program: it invokes the console commands and the analysis programs in order and rebuilds the report.

Table 51: Program inventory. Package programs live under `src/undertow/`, the analysis programs in `scripts/report/`, the configurations in `configs/`, and the report tooling in `report/` and `scripts/`. The analysis programs write the CSV artefacts that the appendix tables and the main-body numbers read.

Program	What it computes
data package	data pipeline: snapshot construction, quality checks and validation reports
simulator package	simulator, exact backtester, parity harness, training and evaluation
dashboard package	live inspection of runs, telemetry and rollouts; not used for any reported result
configuration files	all experiment configurations (pools, splits, grids, costs, training)
`reproduce.sh`	main reproduction program: runs the stages below in order
`analyze_main_run.py`	main joint run: per-window metrics, paired bootstrap intervals, action mix, per-regime tables, decomposition
`static_widths.py`	static one-shot widths on the training and evaluation splits
`extra_benchmarks.py`	the two scripted recentring rules on the evaluation windows
`boundary_sweep.py`	per-window volatility and volume, gas and capital reruns, analytical LVR
`contrast_fee_tier.py`	the full ladder on the 0.30 % pool
`analyze_regime_ablation.py`	the regime-feature ablation arm
`make_appendix_tables.py`	all tables in Appendices E and F, from the CSV artefacts
`make_figures.py`	the main-body figures
`build_report.sh`	builds the report PDF from the LaTeX sources

G.C Code and artefact availability

The complete source code, the tests, the configurations and the LaTeX sources of this report are in the public repository at `https://github.com/Proload1/undertow`; the repository archive contains the code but not the data it reads. The reproduction cannot run from the source alone: the dataset snapshots and the recorded outputs must first be downloaded from the release assets and unpacked over the repository root as shown below. The assets are the validated dataset snapshots at `https://github.com/Proload1/undertow/releases/tag/data-v1` and the recorded evaluation outputs, including the trained checkpoints and training telemetry, at `https://github.com/Proload1/undertow/releases/tag/results-v1`. They are excluded from version control because of their size. The archives unpack over the repository root:

```
tar -xzf eth_usdc_500-data-v1.tar.gz -C data/datasets
tar -xzf eth_usdc_3000-data-v1.tar.gz -C data/datasets
tar -xzf results-v1.tar.gz -C .
```

The primary pool's snapshot is approximately 812 MB and the contrast pool's approximately 293 MB, partitioned by stream and block bucket. Each carries a manifest with row counts and block ranges and a validation report; Appendix B summarises both. The reproduction program does not re-fetch the data

because the fetch step needs API credentials; the reported numbers are produced from the released snapshots and the recorded evaluation artefacts under `results/`.

G.D How to run

From the repository root, after the release assets have been unpacked over it, the wrapper reproduces the analyses, tables, figures, PDF and checks:

```
uv sync
# (download and unpack the release assets first; see above)
scripts/report/reproduce.sh            # analyses, tables, figures, PDF, checks
scripts/report/reproduce.sh --train    # the above, after retraining both arms
```

The `--train` stage retrains the learned policy and the ablation arm; it takes several hours on the hardware below. The analyses take under an hour in total, and the report build takes seconds.

G.E Hardware and runtimes

The experiments ran on the author's training machine: eight CPU cores, 30 GB of memory, Linux under WSL2, and a consumer GPU that the training did not use. Reference runtimes measured during the study: one training seed of one million steps takes about 25 to 40 minutes with four parallel environment processes, so the main run across five seeds takes roughly two and a half hours and the ablation arm about three hours, including the validation scoring of each checkpoint. The full evaluation of one configuration over the thirteen windows takes a few minutes; the boundary sweep's two hundred and sixty reruns take under an hour.

G.F Verification

Three checks keep the reproductions honest. The accounting identity of Appendix D is asserted on each replay. The parity harness bounds the distance between the two engines on the recorded tape. The test suite contains more than two thousand three hundred tests covering the data boundary, position mathematics, the exact fee engine, the environment, training, evaluation and the command-line interfaces, at about 95 % line coverage. A clean reproduction therefore reduces to downloading the two release assets and running the wrapper above.